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Near-field Channel Estimation in THz UM-MIMO Using Deep Compressed Sensing

**A Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Electronics and Communication Engineering**

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Declaration

We hereby declare that the project report entitled **Near-field Channel Estimation in THz UM-MIMO Using Deep Compressed Sensing**, submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Electronics and Communication Engineering from Department of Electrical and Electronic Engineering at University of Tripoli, is a bonafide work done by us under the supervision of **Eng. Aya H.A.Alsmoee**.

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that We have adhered to ethical standards of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the university and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other university.

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9-4-2026

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Abstract

As wireless communications transition toward the 6G era, the Terahertz (THz) band has emerged as a critical frontier for achieving ultra-high data rates and massive bandwidth. However, realizing these gains requires precise Channel Estimation (CE) to manage the directional "pencil-beams" of Ultra-Massive MIMO systems, which utilize hundreds of antenna elements. Traditional estimation techniques, such as Least Squares (LS) or classical Compressed Sensing (CS), increasingly struggle with the high dimensionality of these arrays and the complex hardware constraints of hybrid architectures involving limited RF chains and sub-arrays. Furthermore, the proximity of users in THz scenarios necessitates a shift from far-field planar wave assumptions to more complex spherical wave models.

To overcome these limitations, this project proposes a Deep Compressed Sensing (DCS) framework that replaces traditional mathematical sparsity constraints with structural constraints learned by a generative neural network G_θ . By mapping a low-dimensional latent representation z to the high-dimensional channel space, the model implicitly learns the underlying physical manifold of THz channels from data. During the offline training phase, the network is optimized using a composite loss function that balances Normalized Mean Square Error (NMSE) for accuracy with a Restricted Isometry Property (RIP) loss to ensure the measurement matrix Φ preserves essential signal information and avoids the "null space" trap.

Online inference is treated as an optimization problem where gradient descent is used to find the optimal latent vector $\hat{z}$ that minimizes the reconstruction error. Results demonstrate that this deep learning-based approach significantly outperforms classical methods in terms of estimation accuracy and robustness, particularly in sparse THz environments characterized by distinct clusters and rays. By leveraging the learned priors of the environment, the proposed DCS framework enables high-fidelity channel acquisition with fewer pilot signals, paving the way for efficient and scalable 6G-THz communications.

المخلص

مع انتقال الاتصالات اللاسلكية نحو عصر الجيل السادس، برز نطاق ترددات التيراهيرتز كركيزة أساسية لتحقيق معدلات نقل بيانات فائقة ونطاقات ترددية ضخمة. ومع ذلك، فإن الاستفادة القصوى من هذه التقنية تتوقف على الدقة العالية في عملية تقدير القناة، وهي ضرورية لإدارة الحزم الإشعاعية الضيقة والموجهة في أنظمة الهوائيات متعددة المداخل والمخارج فائقة الضخامة التي تستخدم مئات العناصر الهوائية. تواجه تقنيات التقدير التقليدية، مثل المربعات الصغرى أو طرق الاستشعار المضغوط الكلاسيكية، تحديات متزايدة بسبب الأبعاد العالية لهذه المصفوفات الهوائية، بالإضافة إلى القيود المفروضة على الأجهزة والعتاد، مثل محدودية سلاسل الترددات الراديوية والمصفوفات الفرعية. علاوة على ذلك، فإن قرب المستخدمين من محطات البث في سيناريوهات التيراهيرتز يفرض التحول من نماذج الموجات المستوية التقليدية إلى نماذج الموجات الكروية الأكثر تعقيداً.

ولتجاوز هذه العقبات، تقترح هذه الرسالة إطار عمل يعتمد على الاستشعار المضغوط العميق، والذي يستبدل القيود الرياضية التقليدية للصواب بقيود هيكلية تتعلمها شبكة عصبية توليدية. ومن خلال رسم خريطة تربط بين تمثيل كوامن البيانات منخفض الأبعاد وفضاء الإشارة عالي الأبعاد، يتعلم النموذج ضمناً الأنماط الفيزيائية المعقدة لقنوات التيراهيرتز مباشرة من البيانات. وخلال مرحلة التدريب، يتم تحسين الشبكة باستخدام دالة فقد مركبة توازن بين دقة التقدير وبين ضمان الحفاظ على المعلومات الحيوية للإشارة، مما يمنع الوقوع في فخ التقديرات الرياضية المضللة التي لا تعكس الواقع الفيزيائي.

أما في مرحلة الاستدلال المباشر، فتتم صياغة العملية كمسألة تحسين رياضي تستخدم خوارزميات الانحدار لإيجاد أفضل تمثيل رقمي يقلل من خطأ إعادة بناء الإشارة. وقد أظهرت النتائج أن هذا النهج المعتمد على التعلم العميق يتفوق بشكل ملموس على الطرق الكلاسيكية من حيث دقة التقدير والمرونة، خاصة في بيئات التيراهيرتز التي تتميز بمسارات إشعاعية محددة. ومن خلال الاستفادة من المعرفة المسبقة التي اكتسبتها الشبكة عن البيئة المحيطة، يتيح هذا الإطار المقترح الحصول على تقدير عالي الجودة للقناة باستخدام عدد أقل من إشارات الاختبار، مما يسهل الطريق لاتصالات جيل سادس أكثر كفاءة وقابلية للتوسع في نطاق التيراهيرتز.

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List of Abbreviations

1G First Generation

2D Second Dimension

2G Second Generation

3D Third Dimension

3G Third Generation

4G Fourth Generation

5G Fifth Generation

6G Sixth Generation

AE Antenna Element

AMP Approximate Message Passing

ANN Artificial Neural Network

AoA Angle of Arrival

AoD Angle of Departure

AoSA Array-of-Subarrays

AMP Approximate Message Passing

AR Achievable Rate

BER Bit Error Rate

BS Base Station

CE Channel Estimation

CS Compressed Sensing

CSI Channel State Information

DCNN Deep Convolutional Neural Network

DCS Deep Compressed Sensing

DKL Deep Kernel Learning

DL Deep Learning

DoF Degrees of Freedom

DR Dictionary Reduction

EAR Effective Achievable Rate

ELAA Extremely Large-Aperture Arrays

EM Electromagnetic

eMBB Enhanced Mobile Broadband

FL Federated Learning

FLOPs Floating-Point Operations

FSPL Free-Space Path Loss

FT Fourier Transform

GAN Generative Adversarial Network

GMMV Generalized Multiple Measurement Vector

GSM Global System for Mobile Communications

GSOMP Generalized Simultaneous Orthogonal Matching Pursuit

HITRAN High-Resolution Transmission Molecular Absorption

IoE Internet of Everything

IoT Internet of Things

LDM Location-Division Multiplexing

Leaky-ReLU Leaky Rectified Linear Unit

LMMSE Linear Minimum Mean Square Error

LoS Line-of-Sight

LS Least Squares

LTE Long-Term Evolution

MAML Model-Agnostic Meta-Learning

MIMO Multiple-Input Multiple-Output

MISO Multiple-Input Single-Output

MLP Multilayer Perceptron

mMTC Massive Machine-Type Communications

NLE Non-Linear Estimator

NLoS Non-Line-of-Sight

NMSE Normalized Mean Square Error

OFDM Orthogonal Frequency Division Multiplexing

OMP Orthogonal Matching Pursuit

OTFS Orthogonal Time Frequency Space

PA Power Amplifiers

P-SOMP Polar-domain Simultaneous Orthogonal Matching Pursuit

PS Phase Shifters

PWA Planar-Wavefront Approximation

QAM Quadrature Amplitude Modulation

RF Radio Frequency

RIP Restricted Isometry Property

Rx Receiver

S-V Saleh-Valenzuela

SA Sub-Array

SE Spectral Efficiency

SIMO Single-Input Multiple-Output

SMS Short Message Service

SNR Signal-to-Noise Ratio

SOMP Simultaneous Orthogonal Matching Pursuit

THz Terahertz

Tx Transmitter

UAV Unmanned Aerial Vehicle

UE User Equipment

UM-MIMO Ultra-Massive Multi-Input Multi-output

UPA Uniform Planar Array

URLLC Ultra-Reliable Low-Latency Communications

WCDMA Wideband Code Division Multiple Access

WGAN-GP Wasserstein Generative Adversarial Network with Gradient Penalty

Chapter 1

Introduction

This chapter introduces the project by establishing the background on Terahertz (THz) communications and the evolution of channel estimation techniques toward the Deep Compressed Sensing (DCS) architecture as a solution to the demands of sixth-generation (6G) systems. It outlines the core challenges of Ultra-Massive Multiple-Input Multiple-output (UM-MIMO) at 300 GHz, specifically the complexities introduced by spherical wave propagation and hardware limitations such as restricted Radio Frequency (RF) chains and sub-array (SA) configurations.

1.1 Background

The implementation of THz communications is a cornerstone for the forthcoming 6G networks, designed to meet the unprecedented demands for terabit-per-second data rates and ultra-low latency. Central to this evolution is the deployment of UM-MIMO systems, which utilize hundreds of antenna elements to overcome the severe path loss inherent at THz band through highly directional beamforming.

However, the operational success of these systems is critically dependent on the acquisition of high-fidelity Channel Estimation (CE). In the THz regime, traditional estimation techniques face significant hurdles due to the high dimensionality of the antenna arrays, the hardware constraints of hybrid architectures with limited RF chains, and the transition from planar to spherical wave propagation in the near-field.

To address these complexities, recent research has introduced Deep Compressed Sensing (DCS) as a transformative solution. Unlike classical methods that rely on hand-crafted mathematical sparsity, DCS leverages generative neural networks to learn the underlying structural

manifold of the wireless environment. By enforcing physical constraints such as the Restricted Isometry Property (RIP) during training, this approach optimizes the reconstruction process to achieve superior accuracy even with a reduced number of pilot signals. Consequently, this project investigates the performance benefits of a DCS framework tailored for UM-MIMO architectures, analyzing how deep learning-driven estimation can yield significant improvements in Normalized Mean Square Error (NMSE) and overall Spectral Efficiency (SE) for future 6G systems.

1.2 Related work

This section presents state-of-the-art CE schemes considering both traditional methods and emerging deep learning (DL)-based techniques.

1.2.1 Conventional methods

Beyond the conventional least squares (LS) and linear minimum mean square error (LMMSE) solutions, traditional CE methods involve dictionary-based Compressed Sensing (CS). More precisely, those methods assume that angle of arrival (AoA) and angle of departure (AoD) are taken from a fixed grid and exploit the channel sparsity in angular domains, where CS algorithms such as orthogonal matching pursuit (OMP) [1], and approximate message passing (AMP) [2], can be utilized. Moreover, the authors in [3] propose a generalized simultaneous OMP (GSOMP) algorithm, a variant of the traditional OMP algorithm, by exploiting the common support property and considering the CE problem as a generalized multiple measurement vector (GMMV) problem, where multiple sensing matrices are employed. To overcome the off-grid problem where the actual AoA and AoD deviate from the grid, in [4], the estimates of the angles obtained from GSOMP is refined using electromagnetic (EM)-based methods. Due to the reduced communication distance, several works are tailored for cross-field CE problems. In [5], a simultaneous OMP (SOMP)-based algorithm with dictionary reduction is proposed to solve the cross-field CE problem with lower complexity. In [6], the authors propose an on-grid polar-domain SOMP (P-SOMP) algorithm to estimate channels in the near field effectively. To accommodate the model mismatch for the line-of-sight (LoS) path in the near field, in [7], a LoS-NLoS-separated CE algorithm is proposed where the former is based on parameter estimation and the latter on P-SOMP. LoS sensingbased CE in unmanned aerial vehicle (UAV)-

assisted system is also investigated in [8]. Conventional dictionary-based methods rely heavily on accurate channel modeling and dictionary building. Methods are being developed by either using more accurate channel models or refining the dictionary to mitigate the model mismatch problems. These methods may not perform as well as in the lower frequency band, since the antenna dimension goes higher in the THz band, where the dictionary becomes larger and the computation load is much heavier.

1.2.2 Deep-Learning based methods

The success of DL in various fields makes it a promising candidate for Multiple-Input Multiple-Output (MIMO) CE. DL-based CE algorithms can be categorized into two classes: data-driven methods and model-driven methods. Data-driven approaches, also known as black-box methods, aim to provide an end-to-end mapping from the received signal to the full channel matrix or its corresponding parameters. The authors in [9] utilize the deep kernel learning (DKL) algorithm by using Gaussian process regression, where second-order statistics are learned using a multilayer perceptron (MLP) neural network. Moreover, deep convolutional neural network (DCNN) is extensively exploited in the literature [10], [11], [12], [13] since channel matrices are analogous to pixel-based images, whose intrinsic properties can be captured by convolutional layers. In [10], the channel is initially coarsely estimated using AMP and is then fed to a DCNN as refinement. In addition, insignificant neurons are pruned to reduce inference complexity. In [11], the authors propose a federated learning (FL)-based framework for CE to reduce communication overhead. The works mentioned above leverage the far-field dictionary, which results in high estimation errors in the near-field. To tackle the near-field CE problem, the channel matrix is built first upon the planar wave assumptions in [12]. Then it is multiplied by a correction matrix representing the phase difference between planar and spherical waves. In [13], the authors create a near-field dictionary with an additional dimension for distance, creating a denser grid with higher complexity. Apart from the DCNN structures, generative adversarial network (GAN)-based CE schemes have emerged in recent years [14], [15], [16], [17]. The authors in [14] propose a GAN-based CE framework for wideband channels. GAN is trained to learn the channel distribution from data and generate channel samples from the learned distribution. Then, gradient descent and its variants are used to find the best-generated channel based on the measurements. The authors in [15] and [16] employ the Wasserstein GAN with gradient penalty (WGAN-GP) algorithm to stabilize

training and proposing an FL-based algorithm that distributes the training to multiple users, respectively. In [17], the authors propose a score-based generative model to avoid adversarial training, relying on less stringent assumptions regarding the low-dimensional characteristics of wireless channels. DL-based CE in orthogonal frequency-division multiplexing (OFDM) and orthogonal time frequency space (OTFS) are studied in [18] and [19], respectively. In contrast to the data-driven approaches mentioned above, model-driven methods intertwine DL with domain knowledge by expanding traditional algorithms and substituting non-linear estimators (NLEs) with a neural network. The initial work of deep unfolding methods truncates a set of algorithms to a fixed number of layers, imposing instability and unaffordable training costs [20], [21]. In [22], a joint data-driven and model-driven CE approach is investigated under imperfect hardware consideration. Despite all the merits of DL-based methods, current DL-based estimators often prioritize estimation accuracy while neglecting computational complexity, which is critical in practical implementation. In this paper, the aim is to maintain high estimation accuracy while minimizing the inference complexity by using a similar principle of meta-learning but with only one task dataset [23].

1.3 Project Objectives

The project objectives are summarized as follows:

- To conduct a review of CE technology, with a particular focus on the principles and advantages of Compressive sensing technique.
- To implement a baseline near-field compressed sensing-based channel estimation algorithm for THz UM-MIMO systems.
- To utilize a Deep-learning algorithm to address the high-dimensional challenges of near-field channel estimation.
- To compare the conventional and Deep-learning methods in terms of NMSE, Bit Error Rate (BER), Effective Achievable Rate (EAR) and compression ratio.

1.4 Project Organization

This project has been organized as follows:

- **Chapter 2** establishes the theoretical motivation for the project by tracing the evolution of cellular networks. It details the principles of CE and introduces CS, which is the core mathematical concept used to handle the high-dimensional data of UM-MIMO systems.
- **Chapter 3** defines the mathematical model for a multi-carrier THz UM-MIMO communication system using a sub-connected Array-of-Subarrays (AoSA) architecture. It formulates the channel estimation problem and identifies the limitations of conventional dictionary-based methods. Finally, it introduces the proposed DCS framework and the Model-Agnostic Meta-Learning (MAML)-enhanced algorithm for offline training and online inference.
- **Chapter 4** presents the simulation results and benchmarks the proposed DCS framework against conventional methods. It evaluates the system using metrics such as NMSE, BER, and EAR.
- **Chapter 5** concludes the project with a summary of the key findings and contributions, and suggests potential directions for future work.

Chapter 2

Wireless Networks Evolution and the Foundations of Channel Estimation

This chapter provides the theoretical background for the project. It begins with the evolution of cellular networks from first generation (1G) to 6G to establish the necessary motivation for future systems. The chapter then details the principles of CE and concludes by introducing the Compressed sensing concept, which is central to the models developed in this project.

2.1 Evolution of Cellular Networks from 1G to 6G

The trajectory of mobile communications has been defined by a series of transformative milestones, each fundamentally expanding the functional scope of wireless connectivity. As depicted in Figure 2.1, this progression began in the 1980s with the 1G. While 1G pioneered cellular mobility, it was restricted to analog voice services and characterized by low SE and significant security vulnerabilities.

The 1990s heralded a pivotal transition to the second generation (2G) through the adoption of the Global System for Mobile Communications (GSM) standard. This shift from analog to digital modulation was a landmark achievement, introducing robust signal encryption and enabling the first packet-switched data services, such as the Short Message Service (SMS).

As the requirement for higher data throughput intensified, the third generation (3G) emerged in the early 2000s. By leveraging Wideband Code Division Multiple Access (WCDMA), 3G architectures supported the proliferation of mobile internet access and real-time multimedia services, achieving data rates of up to 2 Mbps.

A subsequent paradigm shift occurred with the fourth generation (4G) and the Long-Term Evolution (LTE) standard. 4G redefined the mobile experience by providing true high-speed broadband through the integration of sophisticated physical layer technologies. Most notably, the implementation of OFDM and MIMO antenna systems laid the technical foundation for the massive-scale spatial multiplexing required by contemporary and future networks. With speeds up to 1 Gbps, 4G facilitated the widespread adoption of smartphones and data-intensive applications. The contemporary deployment of fifth-generation (5G) technology represents

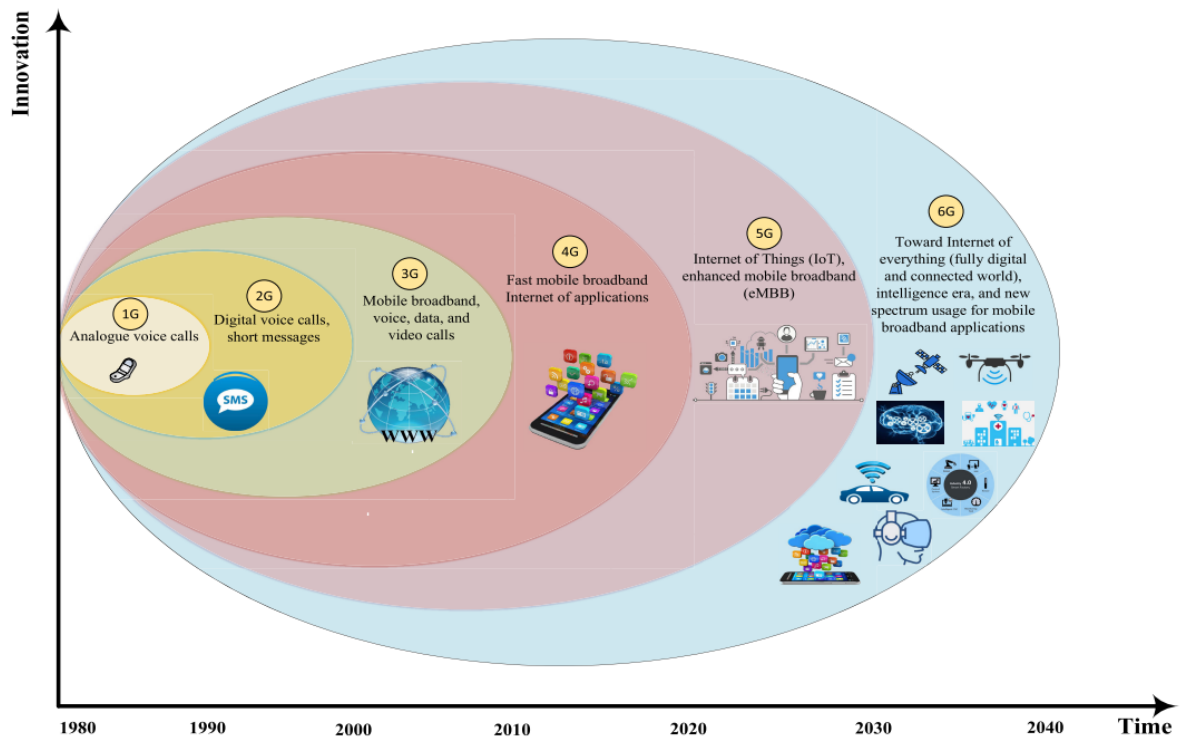


Figure 2.1: The evolution of mobile generations from 1G to 6G [24].

a transformative milestone in wireless standards. 5G was engineered not only to advance enhanced mobile broadband (eMBB) achieving peak throughput in the proximity of 20 Gbps but also to orchestrate a diverse ecosystem of interconnected devices. Through the integration of massive machine-type communications (mMTC) for the Internet of Things (IoT) and ultra-reliable low-latency communications (URLLC) for mission-critical operations, 5G has expanded the scope of cellular networks. Nevertheless, the relentless surge in global data traffic and the increasingly stringent demands of emerging applications suggest that even these advanced capabilities will soon reach their saturation point, necessitating the next frontier of wireless innovation [24].

The conceptualization of sixth-generation (6G) networks is driven by a mandate to support a

sophisticated suite of applications that transcend the current limitations of 5G. The ultimate objective is to realize a natively digital, intelligent, and hyper-connected global landscape. To facilitate this, 6G targets a monumental escalation in performance benchmarks, including: Peak Data Rates: Exceeding 100 Gbps. Latency: Sub-millisecond end-to-end response times (< 1 ms). Reliability: Near-perfect transmission success rates. This "ultra-connectivity" era will serve as the fundamental backbone for truly immersive technologies, such as holographic teleportation and the tactile internet, while simultaneously supporting the dense expansion of the Internet of Everything (IoE). A pivotal shift in the 6G philosophy is the move from a reactive approach—where the system simply adapts to the wireless channel—to a proactive one. By actively controlling the propagation medium to create a smart radio environment, 6G seeks to transform the physical channel into a programmable and intelligent resource, as summarized in Figure 2.2.

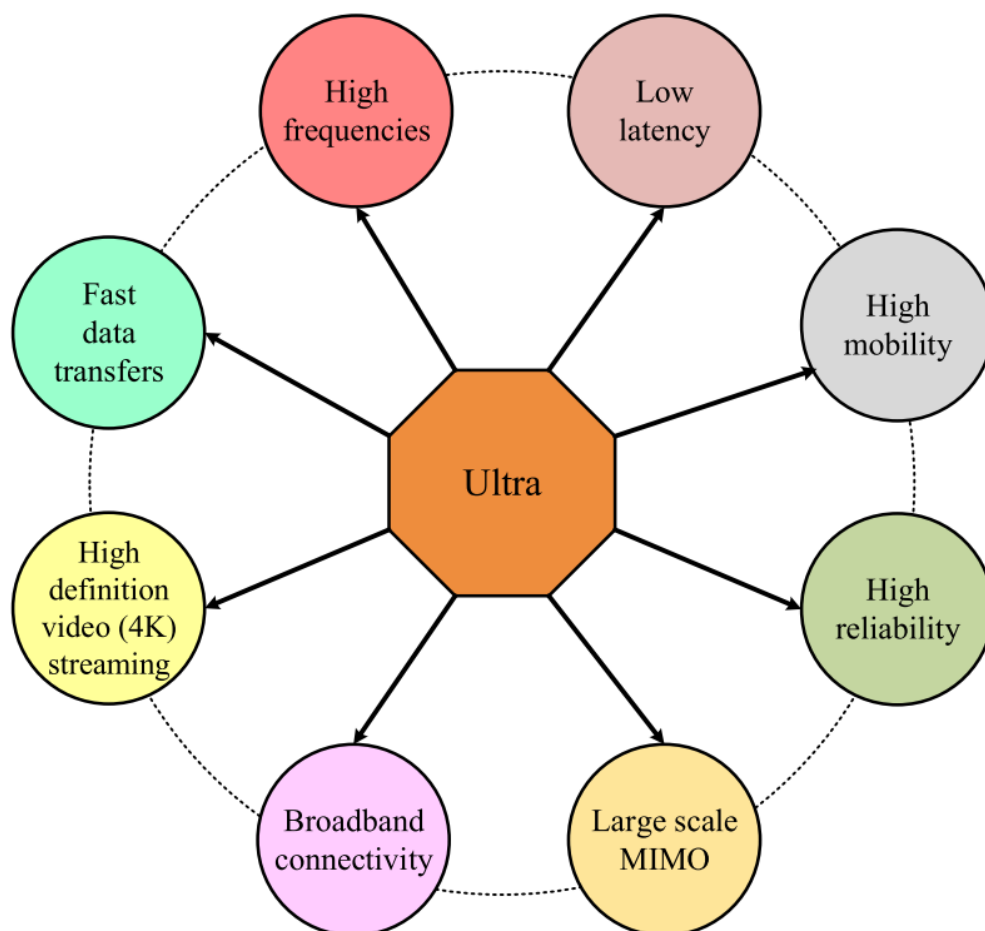


Figure 2.2: A diagram shows the ultra era in 6G networks [24].

2.2 Multiple-Input Multiple-Output (MIMO) Systems

MIMO employs multiple antennas at both transmitter and receiver to exploit spatial diversity, enable spatial multiplexing, and realize coherent beamforming for coverage and interference control. A narrowband link is often expressed as

$$y = Hx + n \quad (2.1)$$

with sufficient rank and channel state information (CSI), parallel streams share the same time-frequency resource while coherent array gain improves reliability. Before outlining the advantages, it is useful to visualize the mechanism. Multiple elements at both ends form several spatial paths; with suitable CSI, the transceiver combines paths for robustness or separates them to carry parallel streams, which is conceptually illustrated in Figure 2.3 [25].

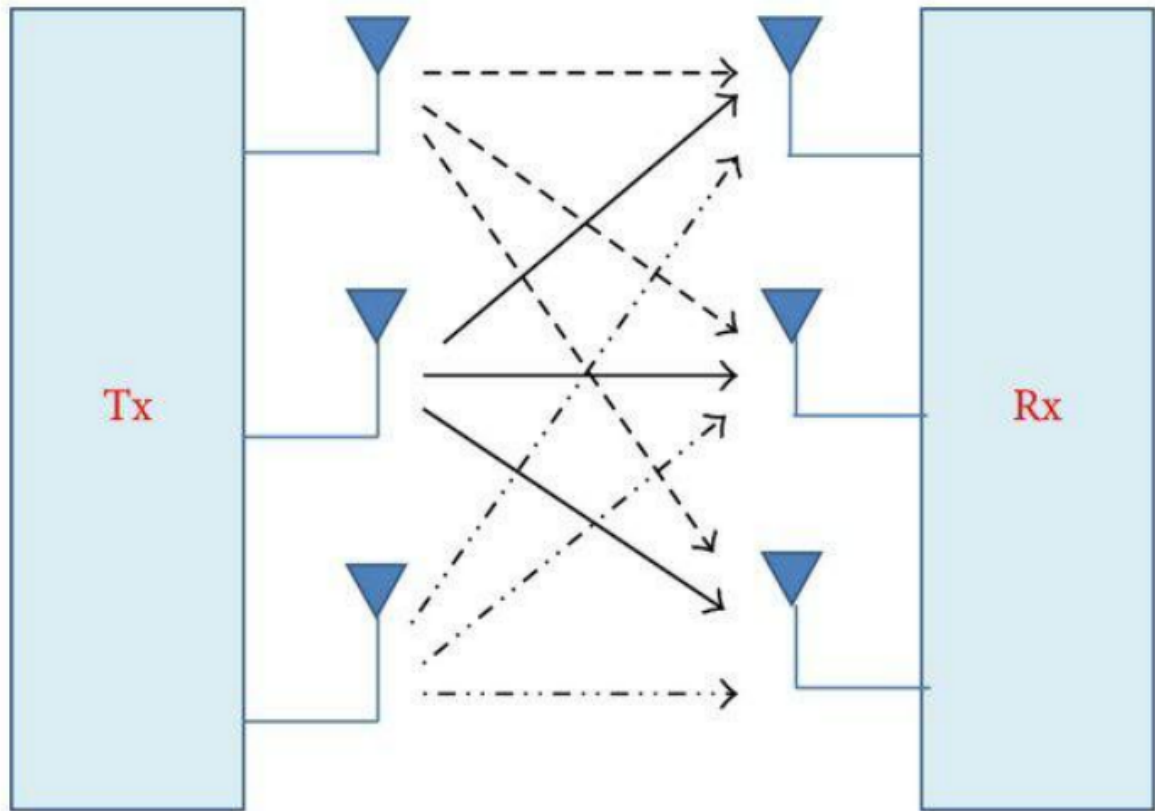


Figure 2.3: Basic MIMO concept with multi-antenna transmitter and receiver [25].

2.3 Ultra-Massive Multiple-Input Multiple-Output Systems

UM-MIMO enlarges the antenna aperture so that many practical links fall within the electromagnetic near-field of the array. In this regime, spherical-wave propagation replaces planar models; channels depend jointly on angle and distance, per-element amplitude/phase vary across the aperture, and visibility can become non-stationary along the array. These effects enable fine spatial focusing and additional degrees of freedom, but they also demand new channel models, estimators, and range–angle codebooks, as demonstrated by the UM-MIMO architecture in Figure 2.4 [26].

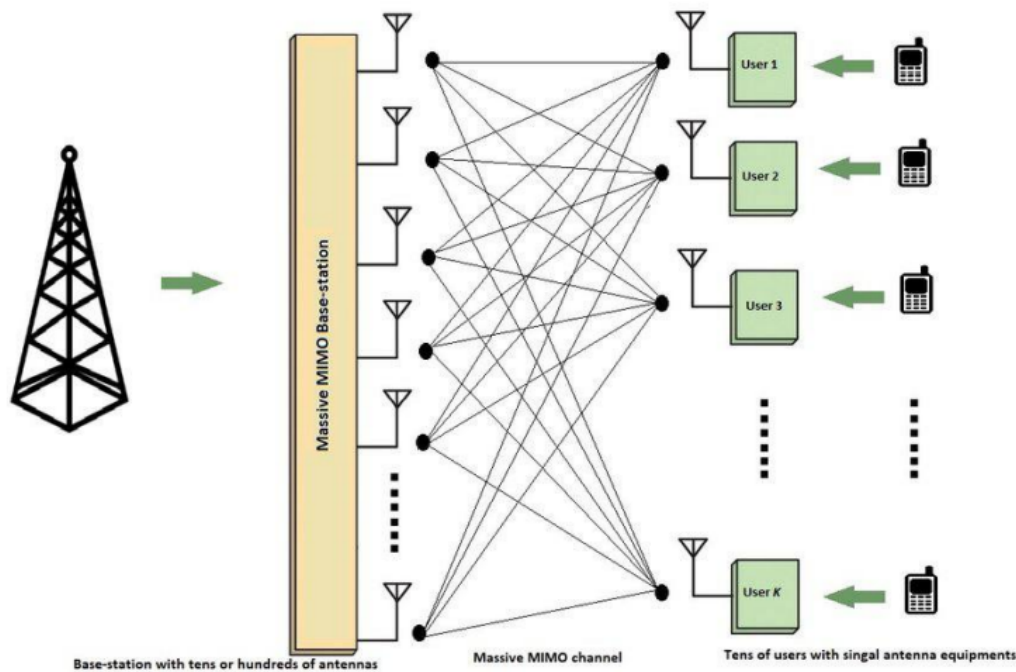


Figure 2.4: UM-MIMO concept-very large array serving many users simultaneously [27].

2.3.1 Advantages of UM-MIMO Systems

The deployment of Extremely Large Aperture Arrays (ELAA) in 6G-class systems is driven by several fundamental operational benefits, particularly in the prevalent near-field regime:

- **Enhanced Spatial Degrees of Freedom (DoF):** The transition to near-field propagation facilitates joint angle-range separability. This enable the implementation of Location-Division Multiplexing (LDM), allowing for more granular interference control and increased network capacity [28].

- **High Coherent Array Gain and Precise Beamfocusing:** ELAA systems provide superior spatial focusing capabilities compared to traditional beamsteering. This results in enhanced signal-to-noise ratio (SNR) for cell-edge users and improved robustness against the severe shadowing and blockage effects characteristic of high-frequency bands[28].
- **Architectural Versatility:** The near-field paradigm is compatible with non-linear array geometries—such as circular, cylindrical, or concentric-ring configurations—and specialized codebooks that sample both the angular and depth dimensions of the channel [28].

2.3.2 Technical Challenges and Implementation Constraints

Despite their performance potential, ELAA and UM-MIMO systems introduce significant modeling and hardware-centric burdens:

- **Estimation and Calibration Complexity:** Modeling spherical-wave propagation introduces complex phenomena such as spatial non-stationarity, depth-of-focus resolution, and mutual coupling between densely packed elements. Consequently, these systems face increased pilot overhead and feedback requirements to maintain accurate CSI.[28]
- **Hardware and System Overheads:** Managing exceptionally large apertures under strict RF-chain constraints necessitates the use of hybrid precoding architectures. This leads to critical design trade-offs involving system dimensionality, computational latency, and stringent power and thermal management budgets.[28]

2.4 THz UM-MIMO System and Channel Modeling

The transition to the THz band for UM-MIMO systems introduces three fundamental physical and hardware-centric challenges. These constraints—ranging from energy bottlenecks to complex wavefront geometries—directly necessitate the development of robust channel estimation frameworks.

2.4.1 Terahertz Band Propagation and Path Loss

Free Space Path Loss

Free-space path loss (FSPL) is the deterministic attenuation that arises purely from spherical spreading of the wavefront along a line-of-sight path in an obstacle-free medium. Starting from the Friis transmission relation, the FSPL in decibels is

$$FSPL(dB) = 32.45 + 20\log_{10}(d) + 20\log_{10}(f) \quad (2.2)$$

where distance d is in kilometers and carrier frequency f in megahertz. This represents the theoretical minimum path loss; additional real-world mechanisms, contribute extra attenuation. Accordingly, FSPL provides the baseline in link-budget computations and guides transmit-power, margin, and antenna-gain selections [29].

Multipath Fading

In practical environments, the received waveform is a superposition of multiple replicas that have undergone reflection, diffraction, and scattering along distinct paths. The differing delays and phases of these components cause rapid fluctuations in amplitude and phase “fading” over small temporal scales. Depending on signal bandwidth and delay spread, fading can be categorized as flat (the channel is approximately constant across the signal band) or frequency-selective (different spectral components experience different gains/phases). Modern systems mitigate or exploit multipath through equalization, multicarrier OFDM, diversity, and MIMO processing to enhance link reliability in rich-scattering environments [29].

2.4.2 Power Consumption in Large-Scale Arrays

The deployment of UM-MIMO systems involving an extensive number of antenna elements frequently exceeding 1024 presents a significant energy efficiency challenge. This prohibitive power consumption arises from two primary factors:

Component-Level Intensity: Individual THz hardware components, such as power amplifiers (PA) and phase shifters (PS), exhibit high power demands, consuming approximately 60 mW and 42 mW respectively.

Array Scale: The sheer volume of these components required for a 1024+ element array

leads to total system consumption reaching several tens of watts. To mitigate these costs, hybrid beamforming architectures are employed to reduce the number of active RF chains. However, this hardware reduction complicates the CE process, as the system must reconstruct a high-dimensional channel from a significantly limited number of digital observations [30].

2.4.3 The Wideband Beam Squint Effect

Unlike traditional microwave systems characterized by narrow fractional bandwidths (e.g., 0.83% at 2.4 GHz), THz communications operate with exceptionally large fractional bandwidths, often reaching 16.7% at a 300 GHz carrier frequency. This wideband nature triggers the beam squint problem:

Frequency Independence: Most hybrid architectures rely on analog phase shifters, which are frequency-independent devices. While they can provide the "correct" phase shift for the central frequency, they induce a phase error for all other subcarriers in a wideband signal. **Gain Degradation:** This mismatch causes the generated beams to "squint" or shift spatially across the frequency band, potentially reducing array gain by up to 5 dB. This implies that the channel matrix is not only spatially sparse but also frequency-selective, requiring an estimation framework capable of compensating for these squint-induced distortions to maintain beamforming precision [30].

2.4.4 Transition to Spherical-Wave Propagation (Near-Field)

A pivotal shift in THz modeling is the invalidation of the traditional Planar-Wavefront Approximation (PWA). In microwave and mmWave systems, the Rayleigh distance (the boundary between near-field and far-field) is typically negligible. However, the Rayleigh distance scales with the square of the array size and the frequency:

Scaling Thresholds: For a 0.1 m array, the Rayleigh distance expands from a mere 4 m at 60 GHz to 67 m at 1 THz.

Spherical Geometry: Given that THz communication distances are often shorter than this threshold, the propagation must be modeled as a Spherical Wavefront. This shift from 2D angular estimation to 3D angle-and-distance modeling introduces additional degrees of freedom. Consequently, traditional far-field "dictionaries" used in compressed sensing become inadequate, justifying the use of DCS to learn the complex, non-linear manifold of near-field

spherical channels [30].

2.5 Array-of-Subarrays (AoSA) Architecture

An AoSA is a hierarchical antenna architecture widely used in UM-MIMO systems, particularly for THz communications. This structure enables efficient management of a very large number of antenna elements while reducing hardware complexity.

In this architecture, both the transmitter (Tx) and receiver (Rx) employ planar antenna arrays distributed on the Y-Z plane of their local Cartesian coordinate systems, where the origin is located at the center of each device. Instead of forming a single large antenna array, the antennas are grouped into multiple smaller units known as sub-arrays.

The total number of sub-arrays is given by

$$N_{SA} = N_z \times N_y \quad (2.3)$$

where N_z and N_y denote the number of sub-arrays along the Z-axis and Y-axis, respectively.

Each sub-array consists of densely arranged antenna elements (AEs). The number of antenna elements within each sub-array is expressed as

$$N_{AE} = \bar{N}_z \times \bar{N}_y \quad (2.4)$$

where $\bar{N}_z$ and $\bar{N}_y$ represent the number of antenna elements along the Z and Y directions within a sub-array [31].

A key feature of the AoSA architecture is that each sub-array is connected to a dedicated RF chain responsible for baseband-to-RF signal conversion. Consequently, the number of RF chains equals the number of sub-arrays. For the transmitter side, this can be written as

$$N_{RF,t} = N_{SA,t} = N_{z,t} N_{y,t}. \quad (2.5)$$

The total number of transmitter antennas is therefore

$$N_t = N_{z,t} N_{y,t} \bar{N}_{z,t} \bar{N}_{y,t}. \quad (2.6)$$

A similar configuration is employed at the receiver, where the number of RF chains and antenna

elements are denoted by $N_{RF,r}$ and N_r , respectively [31].

2.6 Wavefront Modeling and Near-Field Physics

Compared to Massive MIMO, The transition to ULTRA Massive MIMO not only means a sharp increase in the number of antennas but also results in a fundamental change of the EM characteristics. The EM radiation field can generally be divided into far-field and radiation near-field regions. The boundary between these two regions is determined by the Rayleigh distance, also called the Fraunhofer distance. Rayleigh distance is proportional to the product of the square of array aperture and carrier frequency. Outside the Rayleigh distance, it is the far-field region, where the EM field can be approximately modeled by planar waves. Within the Rayleigh distance, the near-field propagation becomes dominant, where the EM field has to be accurately modeled by spherical waves [32].

2.6.1 Far-Field communications vs. Near-Field communication

A commonly used metric for distinguishing between far-field and near-field propagation regions is the Rayleigh distance, also known as the Fraunhofer distance. This distance defines the boundary that determines how EM waves interacting with an antenna array should be modeled.

When the communication distance between the base station (BS) and the user equipment (UE) exceeds the Rayleigh distance, the UE is considered to be located in the far-field region of the BS. In this case, the electromagnetic waves arriving at the antenna array can be accurately approximated as planar waves [31].

Conversely, when the BS–UE distance is smaller than the Rayleigh distance, the UE lies within the near-field region of the BS. Under this condition, the incoming EM waves must be modeled more precisely as spherical waves due to the significant curvature of the wavefront across the antenna array.

A planar wave can be interpreted as a long-distance approximation of a spherical wave. In far-field conditions, the phase variation of the EM wave across the antenna elements can be approximated by a linear function of the antenna index through a Taylor series expansion. This linear phase variation forms a planar wavefront that depends only on the angle of arrival of the signal. As a result, far-field beamforming techniques are able to steer the beam energy toward a

desired direction regardless of the propagation distance. This mechanism is commonly referred to as *beam steering* [32].

However, the linear phase approximation is insufficient for accurately representing spherical waves in near-field scenarios. In the near-field region, the phase of the EM wave must be derived directly from the geometric relationship between the transmitting source and each antenna element. Consequently, the phase variation becomes a nonlinear function of the antenna index. This nonlinear phase contains information about both the angle and the distance of the propagation path between the BS and UE [32].

By exploiting this additional distance-dependent information, near-field beamforming can concentrate the radiated energy at a specific spatial location. This enables simultaneous control of both the angular and distance domains, allowing the beam energy to be focused at a particular point in space. For this reason, near-field beamforming is often referred to as *beam focusing*.

The fundamental differences between planar wavefronts in the far-field region and spherical wavefronts in the near-field region introduce both challenges and opportunities for wireless communication system design, which are visually compared in Figure 2.5 [32].

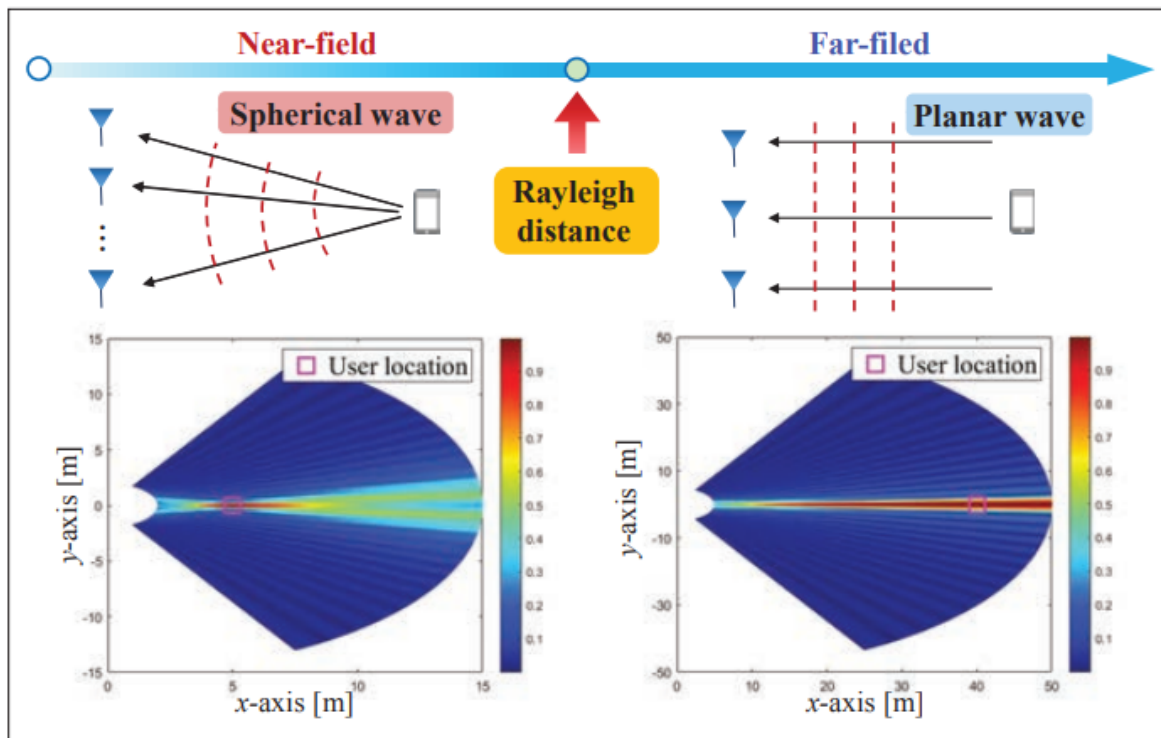


Figure 2.5: Far-field planar wavefront vs. near-field spherical wavefront [32].

2.6.2 Rayleigh Distance and Near-Field Boundary

A fundamental requirement in near-field communications is determining the boundary between the far-field and near-field propagation regions. This boundary is commonly characterized by the Rayleigh distance. In general, the Rayleigh distance is proportional to the square of the antenna array aperture and inversely proportional to the wavelength.

The accurate phase of an EM wave arriving at a BS antenna should be computed based on the spherical wave propagation model. However, in far-field scenarios, this phase is typically approximated using a first-order Taylor expansion, which results in the planar wavefront model. While this approximation simplifies the analysis, it introduces a phase discrepancy that becomes more significant as the communication distance decreases [32].

The Rayleigh distance is defined based on the maximum allowable phase discrepancy caused by this approximation. Specifically, when the largest phase difference between the exact spherical wave model and the planar wave approximation reaches $\pi/8$ across the antenna array, the distance between the centers of the BS and UE arrays is defined as the Rayleigh distance.

If the communication distance is shorter than the Rayleigh distance, the phase discrepancy exceeds $\pi/8$, making the planar wave approximation inaccurate. In such cases, the spherical wave model must be adopted, and the system is considered to operate in the near-field region. Based on this definition, the near-field range varies depending on the antenna configuration of the communication system. In single-input multiple-output (SIMO) and multiple-input single-output (MISO) systems, the near-field boundary is determined by the classical Rayleigh distance, which depends on the square of the base station array aperture.

In MIMO systems, particularly those employing ELAAs at both the transmitter and receiver, the Rayleigh distance is influenced by the apertures of both arrays. Consequently, the near-field region in MIMO systems is proportional to the square of the sum of the base station array aperture and the user equipment array aperture. A summary of these near-field boundaries for typical scenarios is provided in Figure 2.6 [32].

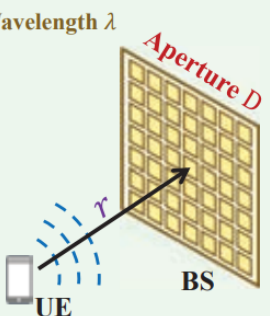
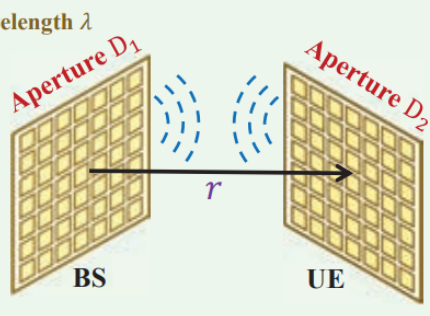
Communication scenarios	SIMO/MISO Wavelength λ 	MIMO Wavelength λ 
	Near-field ranges $r < \frac{2D^2}{\lambda}$	$r < \frac{2(D_1+D_2)^2}{\lambda}$

Figure 2.6: Near-field ranges for typical scenarios [32].

2.7 Compressed Sensing and Sparse Recovery

Traditionally, continuous signal acquisition relies on the Nyquist-Shannon sampling theorem, which dictates that a signal must be sampled at twice its maximum frequency to accurately recover continuous information without aliasing. However, in modern high-bandwidth applications, such as THz communications, this high-speed sampling paradigm consumes immense computational resources and incurs prohibitive hardware costs. To address this, [33] presents Compressed Sensing (CS) as a mathematical framework that circumvents this limitation by synchronizing the sampling and compression processes. This paradigm allows for the recovery of a high-dimensional signal from a minimal number of non-adaptive, linear measurements by exploiting the signal's inherent sparsity [33].

2.7.1 Mathematical Formulation and Sparsity

The fundamental CS problem involves solving an underdetermined linear system. Following the formulation provided by [33], this is expressed as:

$$\mathbf{y} = \Phi \mathbf{x} \quad (2.7)$$

where $\mathbf{x} \in \mathbb{C}^N$ is the original high-dimensional signal, $\mathbf{y} \in \mathbb{C}^M$ is the measurement vector, and $\Phi \in \mathbb{C}^{M \times N}$ is the measurement (or sensing) matrix. By relying on the CS framework, the system captures M valid sampled values rather than the massive N values traditionally required,

ensuring $M \ll N$.

2.7.2 Prerequisites for Exact Recovery

To accurately recover the sparse signal from compressed measurements, the sensing framework must satisfy critical mathematical prerequisites. [34] establishes the Restricted Isometry Property (RIP) as the primary condition for algorithmic stability. According to [34], a measurement matrix Φ satisfies the RIP of order k if there exists a constant $\delta_k \in (0, 1)$ such that for any k -sparse signal $\mathbf{z}$:

$$(1 - \delta_k)\|\mathbf{z}\|_2^2 \leq \|\Phi\mathbf{z}\|_2^2 \leq (1 + \delta_k)\|\mathbf{z}\|_2^2 \quad (2.8)$$

This property ensures that the measurement matrix preserves the Euclidean length of sparse signals, which inherently guarantees exact recovery. Additionally, [33] highlights that the measurement matrix Φ and the sparsifying dictionary Ψ must exhibit high mutual incoherence to ensure that critical, structured information is retained during dimensionality reduction.

2.7.3 Sparse Recovery Algorithms

Recovering the original signal equates to finding the optimal sparse solution to the underdetermined equation. [34] formally defines this as the search for the sparsest solution via ℓ_0 -norm minimization, while noting that this combinatorial search is mathematically NP-hard. To overcome this computational barrier, two primary algorithmic strategies are utilized in the literature:

- **Convex Optimization:** As detailed by [34], these methods relax the non-convex ℓ_0 -norm into a computationally tractable convex problem. Basis Pursuit serves as the classic example, substituting the ℓ_0 -norm with the ℓ_1 -norm to find an approximating signal.
- **Greedy Algorithms:** Alternatively, techniques such as Orthogonal Matching Pursuit (OMP) and Simultaneous Orthogonal Matching Pursuit (SOMP) avoid complex global optimization. [35] utilize these greedy approaches in modern array processing, explaining that they iteratively select the columns (atoms) of the dictionary Ψ that best correlate with the residual of the measurement vector to calculate the sparse coefficients step-by-step.

2.7.4 Limitations in Near-Field THz UM-MIMO Systems

While the foundational texts [36] establish CS as a robust mathematical tool, its direct application to THz Ultra-Massive MIMO systems reveals severe scalability and modeling bottlenecks. [33] specifically address these modern challenges, noting that classical CS must be heavily adapted for these extreme environments. For instance, to manage computational complexity, [33] proposed a dictionary reduction method based on SOMP to estimate wideband near-field THz channels by exploiting spatial sparsity across sub-arrays.

However, the physical properties of UM-MIMO systems actively undermine the fundamental prerequisites of classical CS. Because of the massive array apertures and millimeter-scale wavelengths, THz communications frequently operate in the near-field region, where electromagnetic waves propagate as spherical wavefronts rather than planar waves. This near-field effect, combined with classical dictionaries relying on discrete angular grids, causes the signal's energy to smear across multiple predefined grid points, destroying the assumption of sparsity [33]. When the true continuous channel parameters do not align perfectly with the predefined discrete basis, classical greedy algorithms suffer from severe grid-mismatch errors.

Beyond the modeling errors, the iterative nature of these classical solvers presents a significant temporal bottleneck. As illustrated in the top portion of Figure 2.7, traditional CS reconstruction relies on a feedback loop that repeats until convergence, a process that often exceeds the strict coherence time of THz channels. The transition to the feedforward structure shown in the bottom of Figure 2.7 represents the shift toward the Deep Learning architectures explored in the following sections [37].

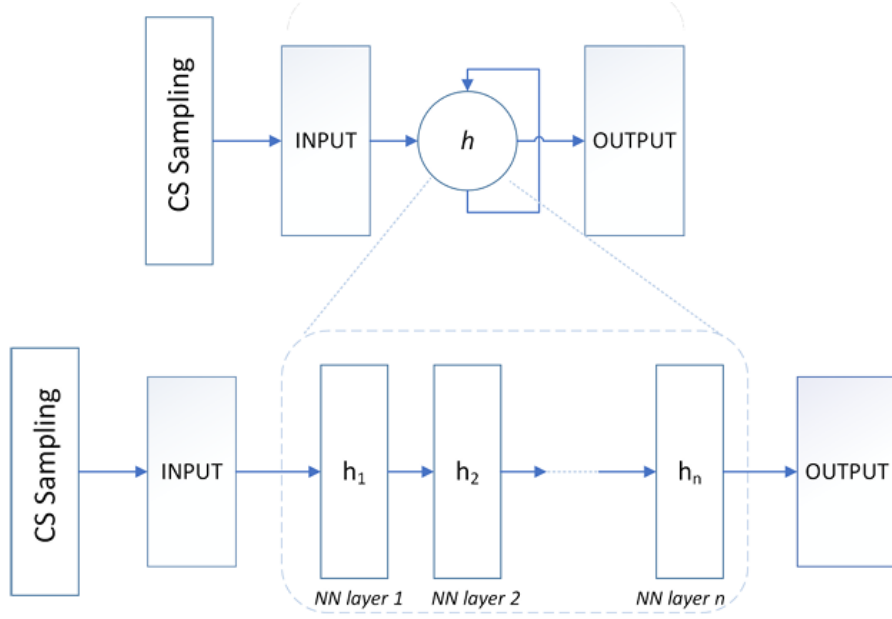


Figure 2.7: The iterative structure of classical reconstruction (top) versus the feedforward unrolled structure (bottom) [37].

2.7.5 The Shift Toward Deep Compressed Sensing (DCS)

To overcome the inherent limitations and grid-mismatch errors of the handcrafted analytical dictionaries described in [35], recent research advocates for a paradigm shift toward Deep Compressed Sensing (DCS). As demonstrated by [38], the DCS framework abandons the strong, often sub-optimal assumption of analytical sparsity. Instead, it replaces the traditional dictionary Ψ with a data-driven structural prior enforced by a pre-trained generative neural network, G_θ . In this model, the signal such as the high-dimensional THz channel is restricted to a low-dimensional manifold learned directly from the data distribution, expressed as $\mathbf{x} \approx G_\theta(\mathbf{z})$, where $\mathbf{z}$ is a compressed latent representation [38].

This alignment between generative modeling and compressed sensing is particularly effective for THz UM-MIMO systems. First, the generative model provides a continuous representation of the channel, effectively circumventing the "off-grid" leakage problems that plague greedy algorithms like SOMP. Second, [38] introduces a meta-learning strategy that significantly accelerates recovery, while [31] adapts this framework to capture the intricate near-field spherical wavefronts inherent to THz frequencies. Consequently, DCS not only enhances recovery accuracy under severe pilot undersampling ($M \ll N$) but also achieves the near-instantaneous estimation speeds required for real-time THz communication.

2.8 Introduction to Deep Learning for Signal Recovery

As established in previous sections, classical CS algorithms face severe computational and representational bottlenecks when applied to high-dimensional problems, such as acquiring CSI in massive antenna arrays. To overcome the limitations of handcrafted analytical dictionaries, modern signal processing has increasingly turned toward Deep Learning (DL). By leveraging large datasets, DL models can automatically discover and extract the underlying non-linear representations of complex wireless environments, offering a powerful alternative to traditional convex optimization and greedy algorithms.

2.8.1 Fundamentals of Deep Neural Networks

At its core, Deep Learning relies on Artificial Neural Networks (ANNs) to approximate complex, non-linear functions mapping inputs to outputs such as mapping a small set of received pilot signals to a full high-dimensional channel matrix. A standard feed-forward neural network consists of multiple layers of interconnected neurons. To avoid notational collision with the channel vector $\mathbf{h}$, let $\mathbf{v}^{(l)}$ denote the intermediate feature representation at layer l . The transformation as the data passes through a hidden layer is mathematically defined as:

$$\mathbf{v}^{(l)} = \sigma(\mathbf{W}^{(l)}\mathbf{v}^{(l-1)} + \mathbf{b}^{(l)}) \quad (2.9)$$

where $\mathbf{W}^{(l)}$ represents the trainable weight matrix, $\mathbf{b}^{(l)}$ is the bias vector, and $\sigma(\cdot)$ is a non-linear activation function, such as the Leaky Rectified Linear Unit (Leaky-ReLU) [39].

Unlike classical algorithms that rely on predefined mathematical rules, a neural network learns these weights and biases during an offline training phase. By defining a loss function that measures the discrepancy between the network's prediction and the true data, the network's parameters are iteratively updated using gradient descent and the backpropagation algorithm. This data-driven optimization allows deep neural networks to learn hierarchical spatial features that are significantly more expressive than the fixed bases used in traditional CS [39].

2.8.2 Generative Models as Structural Priors

While traditional DL models excel at classification and regression, inverse problems like compressed channel estimation require generative capabilities the ability to reconstruct a

massive dimensional signal from a heavily compressed observation.

The conceptual bridge between DL and CS was formalized by [40], who demonstrated that the traditional sparsity constraint in a discrete dictionary (Ψ) could be entirely replaced by a Deep Generative Prior. In this paradigm, a pre-trained generative neural network, denoted as G_θ , is used to map a low-dimensional latent vector $\mathbf{z} \in \mathbb{C}^d$ to the high-dimensional channel space $\mathbf{h} \in \mathbb{C}^N$, where $d \ll N$. Because the generator is trained on a specific dataset (such as simulated or measured multipath channel matrices), it restricts the reconstructed channel to the natural continuous manifold of the data, expressed as:

$$\mathbf{h} \approx G_\theta(\mathbf{z}) \quad (2.10)$$

This effectively eliminates the "off-grid" leakage problems associated with fixed analytical dictionaries [40].

2.8.3 The Deep Compressed Sensing (DCS) Framework

While the introduction of generative priors revolutionized signal recovery, early models suffered from extreme computational latency. Recovering the signal required thousands of gradient descent steps to optimize the latent vector $\mathbf{z}$ for every new measurement, which is entirely incompatible with the strict coherence times of modern wireless channels.

To resolve this bottleneck, [38] introduced the formal DCS framework. As illustrated in Figure 2.8, DCS fundamentally improves upon earlier generative methods by optimizing the latent representation so that the measurement produced by the generator matches the actual compressed observation. Rather than training only a generator to produce realistic signals, the DCS framework learns a latent space and inference strategy that enable rapid recovery from compressed measurements.

When a receiver captures a compressed pilot measurement $\mathbf{y} \in \mathbb{C}^M$ (labeled as $\mathbf{m}$ in Figure 2.8), the DCS recovery algorithm seeks to find the optimal latent vector $\mathbf{z}^*$ (represented by $\hat{\mathbf{z}}$) by minimizing an objective composed of a measurement-fidelity term and a regularization term:

$$\mathcal{L}(\mathbf{z}) = \|\mathbf{y} - \Phi G_\theta(\mathbf{z})\|_2^2 + \lambda \|\mathbf{z}\|_2^2 \quad (2.11)$$

where $\Phi \in \mathbb{C}^{M \times N}$ is the pilot measurement matrix (labeled as $\mathbf{F}$ in the figure), G_θ is the generator with frozen weights θ , and λ is a Tikhonov regularization parameter.

In relation to Figure 2.8, the upper branch represents the formation of the true compressed measurement ($\mathbf{x} \rightarrow \mathbf{m}$), while the lower branch represents the generated signal $G_\theta(\mathbf{z})$ (shown as $\hat{\mathbf{x}}$) being mapped through the same measurement operator to produce an estimated measurement $\hat{\mathbf{m}}$. The latent vector is then iteratively updated indicated by the red optimization arrow until this estimated measurement matches the received one, thereby minimizing the error $\mathbf{E}$.

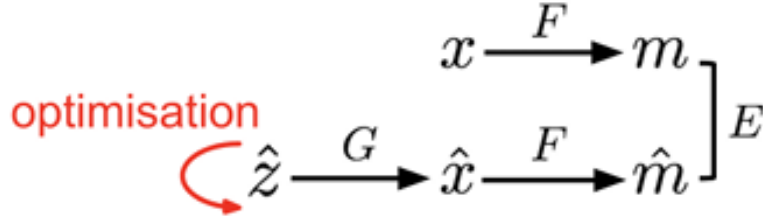


Figure 2.8: The DCS framework, optimizing the latent vector $\hat{\mathbf{z}}$ to minimize measurement error [38].

Because of the meta-learning strategy utilized during training, the DCS framework learns how to navigate the latent space efficiently. Consequently, this optimization requires only a few gradient steps to converge using optimizers like Adam, dramatically accelerating the reconstruction process while maintaining the high fidelity provided by the generative prior [38].

2.8.4 Classification of Deep Learning Architectures

The integration of Deep Learning with Compressed Sensing has evolved rapidly, yielding a diverse landscape of network architectures. As comprehensively surveyed by [37] and mapped in the taxonomy of Figure 2.9, the primary motivation unifying these approaches is the need to alleviate the severe computational bottlenecks of classical iterative recovery algorithms. By shifting the computational burden to an offline training phase, DL-based systems enable the near-instantaneous online inference required for time-sensitive applications, such as estimating channels within ultra-short THz coherence times.

Within this landscape, [37] categorizes DL-CS architectures into two dominant paradigms:

- **Algorithm Unrolling:** These architectures explicitly mimic the iterative steps of classical convex optimization solvers. Each layer of the neural network represents a single iteration of the classical algorithm, with weights optimized via training data rather than mathematical derivation.

- **Direct and Generative Reconstruction:** Unlike unrolled networks, these models do not attempt to replicate traditional mathematical solvers. Instead, they leverage structures like Autoencoders, GANs, or the DCS framework to learn a direct mapping from the compressed measurements to the high-dimensional signal space.

For the complex, non-linear physical manifolds of near-field THz UM-MIMO systems, the Direct/Generative paradigm is highly advantageous. While Algorithm Unrolling remains fundamentally constrained by the classical assumption of sparsity in a fixed basis, generative models bypass this limitation entirely. As utilized in the core methodology of this thesis, the generative approach replaces the restrictive analytical dictionary with a fluid, data-driven prior, providing the flexibility necessary to capture the severe spherical wavefront distortions of the near-field environment [37, 38].

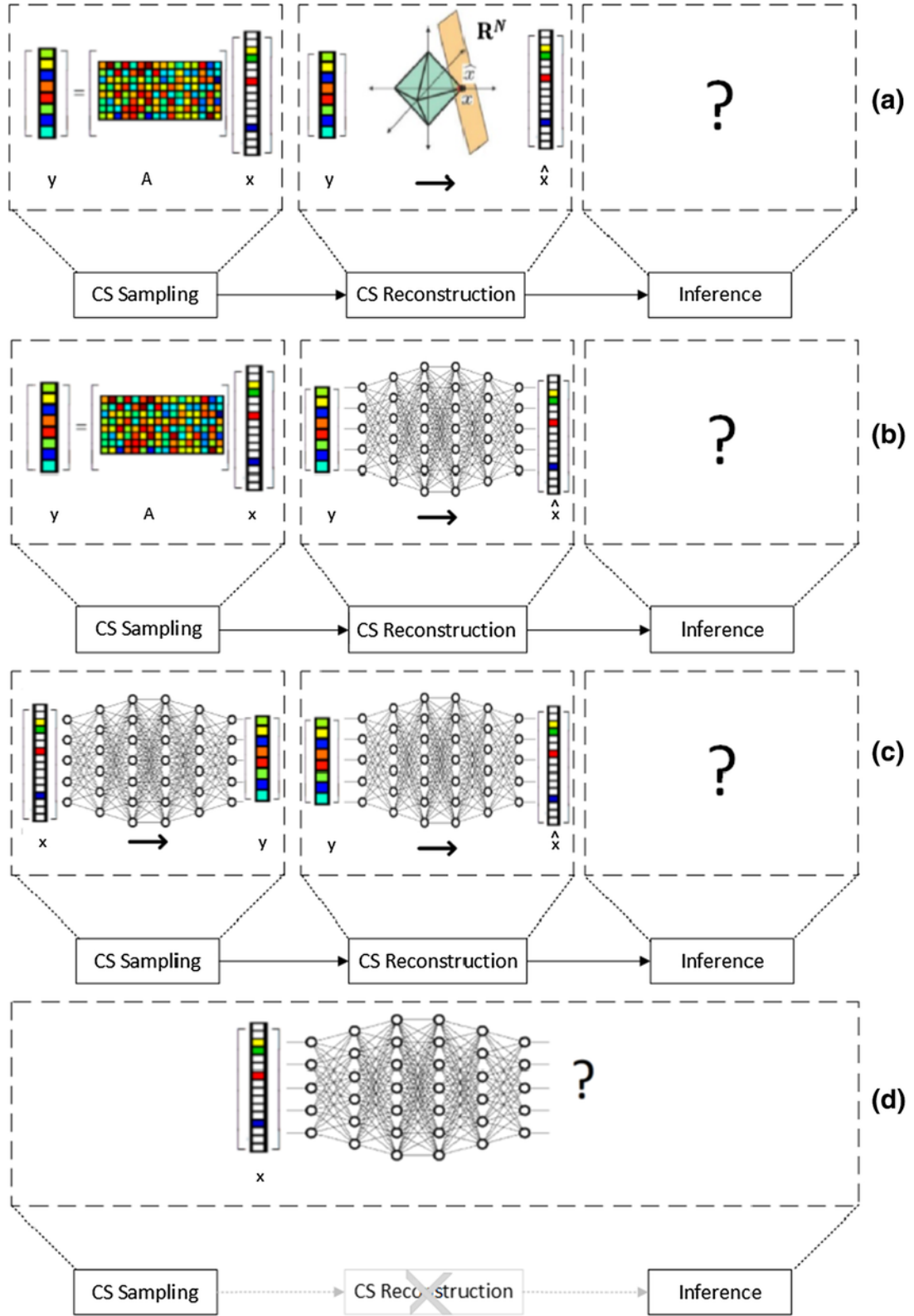


Figure 2.9: Structural classification of Deep Learning integration in Compressed Sensing: a) conventional CS using iterative solvers; b) DL for CS reconstruction; c) End-to-end DL for both CS sampling and reconstruction (the DCS approach); and d) DL for direct inference [37].

Chapter 3

System Model and Proposed DCS Architecture

This chapter establishes the architectural and mathematical foundations for the proposed THz channel estimation framework. First, the physical UM-MIMO transceiver architecture is defined, utilizing a sub-connected AoSA structure [33]. Subsequently, the physical propagation environment is modeled, emphasizing the near-field spherical wavefronts inherent to massive arrays operating at millimeter-scale wavelengths. Finally, the channel estimation task is formulated mathematically, contrasting the limitations of conventional dictionary-based CS with the proposed DCS framework [31].

3.1 Introduction to THz UM-MIMO Channel Estimation

To fully realize the ultra-high data rates promised by THz band communications, UM-MIMO antenna arrays are deployed to overcome the severe free-space path loss inherent at such high frequencies. However, to leverage the massive array gain through precise beamforming and spatial multiplexing, the receiver must possess accurate knowledge of the CSI. Therefore, accurate CE is the fundamental prerequisite for reliable THz communication [31].

Estimating the THz UM-MIMO channel presents two massive challenges. First, the sheer number of antennas creates a prohibitively large channel matrix, resulting in massive pilot overhead and computational complexity. Second, the massive aperture of the array combined with the millimeter/sub-millimeter wavelengths of THz signals means the communication link often operates in the near-field regime, where traditional far-field planar wave assumptions

fail [31, 33]. This chapter details the physical system model and formulates the channel estimation problem, transitioning from classical baseline recovery methods to the proposed DCS framework.

3.2 System Architecture and Transmission Model

This section details the physical hardware configuration and the mathematical formulation of the transmitted and received signals. To mitigate the severe hardware complexity and power consumption typical of Terahertz systems, a hybrid beamforming architecture relying on a sub-connected array topology is established.

3.2.1 Ultra-Massive MIMO with AoSA Structure

A downlink multi-carrier THz UM-MIMO communication system utilizing an ultra-wide bandwidth is considered. Both the Tx and Rx adopt a sub-connected AoSA hybrid beamforming architecture, where each individual SA is exclusively driven by a single RF chain [33].

The Tx is equipped with $N_{SA}^{(t)}$ subarrays, and each SA consists of a uniform planar array (UPA) containing $N_{AE}^{(t)}$ tightly-packed AEs. The total number of transmit antennas is therefore $N_t = N_{SA}^{(t)} \times N_{AE}^{(t)}$. The Rx follows an identical structural convention, resulting in $N_r = N_{SA}^{(r)} \times N_{AE}^{(r)}$ total receive antennas. The complete structural block diagram for this hybrid beamforming architecture is illustrated in Figure 3.1.

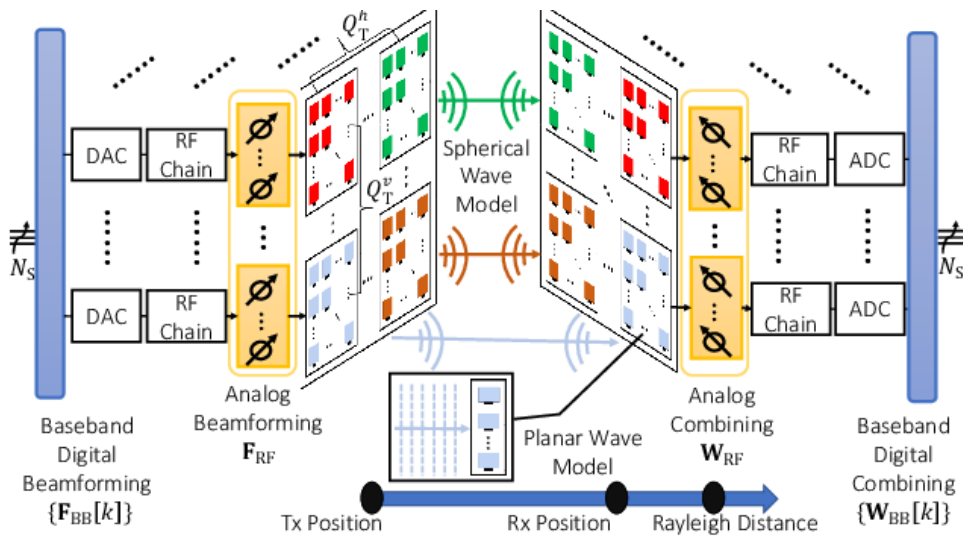


Figure 3.1: Block diagram of the THz UM-MIMO sub-connected AoSA hybrid beamforming architecture [33].

3.2.2 Baseband Transmission Model

Operating over K subcarriers, the complex baseband transmitted signal at the k -th subcarrier is formulated as:

$$x[k] = F_{RF} F_{BB}[k] s[k] \quad (3.1)$$

where $s[k] \in \mathbb{C}^{N_s \times 1}$ represents the vector of data streams. The signal is sequentially processed by a digital baseband precoder $F_{BB}[k]$ and a frequency-flat analog RF precoder F_{RF} . Due to the sub-connected architecture, In the AoSA architecture, the hardware complexity is significantly reduced by mapping each SA to a dedicated RF chain. Due to this partially-connected configuration, the analog beamforming matrix $\mathbf{F}_{RF} \in \mathbb{C}^{N_t \times N_{RF,t}}$ exhibits a sparse, block-diagonal structure. This indicates that each AE is interfaced only with the RF chain associated with its specific sub-array, thereby avoiding the high power consumption and routing overhead of a fully-connected network.[31] Mathematically, $\mathbf{F}_{RF}$ is defined as:

$$\mathbf{F}_{RF} = \begin{bmatrix} \mathbf{f}_1^{RF} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{f}_2^{RF} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{f}_{N_{RF,t}}^{RF} \end{bmatrix} \quad (3.2)$$

where $\mathbf{f}_j^{RF} \in \mathbb{C}^{N_{AE,t} \times 1}$ represents the analog weight vector for the j -th sub-array, for all $j \in \{1, 2, \dots, N_{RF,t}\}$. Given that each AE is connected to a finite-bit phase shifter operating under a constant-modulus constraint, the non-zero elements in $\mathbf{F}_{RF}$ possess uniform magnitude with varying phases. Specifically, the i -th element of the vector $\mathbf{f}_j^{RF}$ is expressed as:

$$f_j^{RF}[i] = \frac{1}{\sqrt{N_{AE,t}}} e^{j\psi_{i,j}}, \quad \forall i \in \{1, 2, \dots, N_{AE,t}\} \quad (3.3)$$

In this formulation, $\psi_{i,j} \in [0, 2\pi]$ denotes the phase shift applied to the i -th antenna element of the j -th sub-array. The normalization factor $1/\sqrt{N_{AE,t}}$ ensures that the total power radiated by each sub-array is maintained at a constant level. This architectural model defines the hardware-limited search space for the channel estimation and beamforming optimization strategies discussed in subsequent sections [31]. After propagating through the frequency-selective THz channel, the received baseband signal at the k -th subcarrier is given by:

$$y[k] = C_{BB}^H[k] C_{RF}^H H[k] x[k] + C_{BB}^H[k] C_{RF}^H n[k] \quad (3.4)$$

where $H[k] \in \mathbb{C}^{N_r \times N_t}$ is the overall frequency-domain channel matrix, $n[k]$ represents the additive Gaussian noise vector, and C_{RF} and $C_{BB}[k]$ are the analog and digital combining matrices, respectively [31].

3.3 Near-Field Spherical Channel Model

The wideband frequency-selective THz channel is characterized by a multi-subcarrier framework, where the total available bandwidth is partitioned into K orthogonal subcarriers. In this model, the channel state at each subcarrier is defined by the superposition of both LoS and Non-Line-of-Sight (NLoS) propagation paths. Considering a discrete-time representation with N_u delay taps in the time domain, the delay- u channel matrix $\mathbf{H}_u$ at the k -th subcarrier frequency f_k is mathematically formulated as:

$$\mathbf{H}_u = \mathbf{H}_u^{LoS} + \mathbf{H}_u^{NLoS} \quad (3.5)$$

where $\mathbf{H}_u^{LoS}$ and $\mathbf{H}_u^{NLoS}$ represent the LoS and NLoS components of the channel response, respectively. This decomposition serves as the basis for the geometric channel model, where the specific spatial and temporal signatures of each path, such as time-of-arrival, angle-of-departure, and angle-of-arrival, are captured to facilitate high-precision channel estimation.

3.3.1 Line of Sight and Non-Line of Sight Propagation

The LoS channel component is modeled under a geometric free-space propagation framework, accounting for the distinct propagation trajectories between each pair of transmit and receive AEs. Let $[\mathbf{H}_u^{LoS}]_{n_r, n_t}$ denote the channel coefficient between the n_r -th receive AE and the n_t -th transmit AE. For a given subcarrier frequency f_k and a discrete-time delay tap u , the LoS response is mathematically formulated as:

$$[\mathbf{H}_u^{LoS}]_{n_r, n_t} = \alpha_0(f_k, d_{n_r, n_t}) \delta(uT_s - \tau_{n_r, n_t}) \quad (3.6)$$

where T_s is the system sampling period and $\delta(\cdot)$ is the Dirac delta function mapping the path to the discrete delay domain. The term d_{n_r, n_t} represents the exact Euclidean distance between the AE pair. Furthermore, $\alpha_0(f_k, d_{n_r, n_t})$ and τ_{n_r, n_t} denote the frequency-dependent path gain and the propagation delay, respectively. This element-wise modeling is essential in the near-field

regime, as it captures the spherical phase curvature across the ultra-massive aperture.

The NLoS component is modeled using the Saleh-Valenzuela (S-V) stochastic framework to account for multipath scattering[41]. The delay-domain NLoS channel matrix $\mathbf{H}_u^{NLoS}$ is expressed as the summation of L discrete propagation paths:

$$\mathbf{H}_u^{NLoS} = \sum_{\ell=1}^L \alpha_{\ell} \mathbf{a}_r \mathbf{a}_t^H \delta(uT_s - \tau_{\ell}) \quad (3.7)$$

where L denotes the total number of NLoS paths, α_{ℓ} is the complex path gain, and τ_{ℓ} represents the associated propagation delay for the ℓ -th path. The vectors $\mathbf{a}_r$ and $\mathbf{a}_t$ are the receive and transmit array response vectors, respectively. Notably, in the UM-MIMO near-field regime, these vectors are functions of both the angles of arrival/departure and the distance from the scatterer to the array aperture.

The transition from the delay domain to the frequency domain is achieved via the Fourier Transform (FT). The frequency-domain channel response $\mathbf{H}[k]$ at the k -th subcarrier is given by:

$$\mathbf{H}[k] = \sum_{u=0}^{N_u-1} (\mathbf{H}_u^{LoS} + \mathbf{H}_u^{NLoS}) e^{-j\frac{2\pi k}{K}u} \quad (3.8)$$

$$= \mathbf{H}^{LoS}[k] + \mathbf{H}^{NLoS}[k] \quad (3.9)$$

By applying the phase shift properties of the Fourier Transform, the frequency-domain NLoS component is formulated as:

$$\mathbf{H}^{NLoS}[k] = \sum_{\ell=1}^L \alpha_{\ell} \mathbf{a}_r \mathbf{a}_t^H e^{-j2\pi \frac{kB}{K} \tau_{\ell}} \quad (3.10)$$

where B represents the total system bandwidth and K is the number of subcarriers. Similarly, the element-wise frequency-domain LoS response is defined as:

$$[\mathbf{H}^{LoS}[k]]_{n_r, n_t} = \alpha_0(f_k, d_{n_r, n_t}) e^{-j2\pi \frac{kB}{K} \tau_{n_r, n_t}} \quad (3.11)$$

This frequency-selective model captures the wideband characteristics of the THz channel, providing the necessary framework for subcarrier-based channel estimation.

3.3.2 Spherical Wavefront Array Response

In the near-field regime, the array response matrices A_T and A_R depend not only on the angles of departure and arrival, but heavily on the absolute distance from the scatterer (or user) to the individual antenna elements.

Following the precise near-field formulation [31], the spatial array response vectors for the transmitter ($\mathbf{a}_t$) and receiver ($\mathbf{a}_r$) are defined by the spherical phase shifts across the array elements:

$$\mathbf{a}_t(\theta_t, \phi_t, d_t) = \frac{1}{\sqrt{N_t}} \left[e^{-j\frac{2\pi}{\lambda}(d_t^{(1)}-d_t)}, \dots, e^{-j\frac{2\pi}{\lambda}(d_t^{(N_t)}-d_t)} \right]^T \quad (3.12)$$

$$\mathbf{a}_r(\theta_r, \phi_r, d_r) = \frac{1}{\sqrt{N_r}} \left[e^{-j\frac{2\pi}{\lambda}(d_r^{(1)}-d_r)}, \dots, e^{-j\frac{2\pi}{\lambda}(d_r^{(N_r)}-d_r)} \right]^T \quad (3.13)$$

where d_t and d_r represent the reference distances from the scatterer to the center of the transmit and receive arrays, respectively, and λ is the signal wavelength. The terms $d_t^{(m)}$ and $d_r^{(m)}$ denote the absolute geometric distances to the m -th individual antenna element.

These absolute element distances are calculated utilizing the directional unit vectors derived from the AoD and AoA. The 3D local coordinate system defining these spatial angles relative to the physical antenna array is illustrated in Figure 3.2.

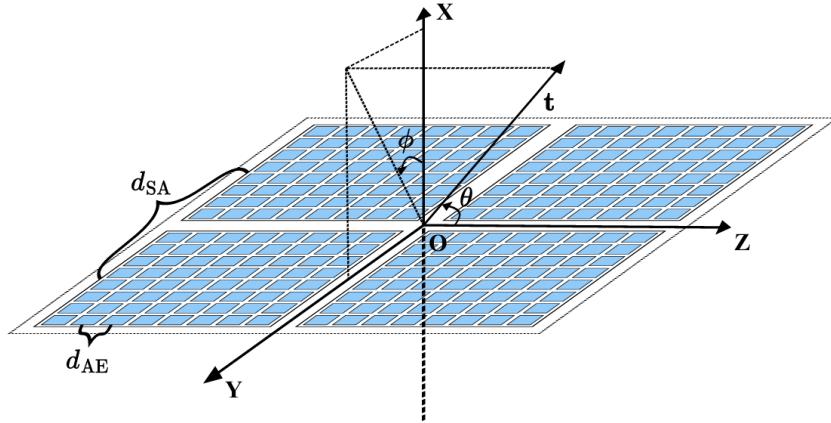


Figure 3.2: The 3D local coordinate system illustrating the azimuth (ϕ) and elevation (θ) angles relative to the planar antenna array [31].

3.4 Compressed Sensing Formulation and Limitations

This section translates the physical received signal into a mathematical estimation problem. First the linear observation model used for pilot-based training is constructed, and then

analyze why classical dictionary-based recovery techniques suffer from critical performance degradation in near-field scenarios.

3.4.1 Underdetermined Linear Observation Model

It is assumed that the coherence time of the channel significantly exceeds the duration of a single symbol, allowing the communication process to be distinctly separated into a training phase and a subsequent data transmission phase. This premise holds true since THz symbol durations operate on a picosecond scale, whereas the channel's coherence time extends into the millisecond range [31].

For the sake of notational simplicity, the subcarrier index is temporarily omitted. Throughout the training phase, the transmitter dispatches a sequence of N_p pilot signals, which the receiver utilizes to estimate the underlying channel matrix. Applying distinct precoders and combiners for beam training, the received signal for the m -th training symbol is formulated as:

$$y_m = C_m^H H x_m + C_m^H n_m \quad (3.14)$$

where C_m denotes the combining matrix, x_m is the transmitted pilot vector, and n_m is the additive noise.

To convert this into a standard Compressed Sensing format, the channel matrix must be vectorized. This relies on the fundamental matrix vectorization properties:

$$\text{vec}(AC) = (I \otimes A)\text{vec}(C) \quad (3.15)$$

$$\text{vec}(AXC) = (C^T \otimes A)\text{vec}(X) \quad (3.16)$$

Applying these properties to our received signal, the linear system for the m -th symbol becomes:

$$y_m = (x_m^T \otimes C_m^H)h + C_m^H n_m = \Phi_m h + C_m^H n_m \quad (3.17)$$

where $h = \text{vec}(H)$ is the vectorized physical channel, and $\Phi_m = (x_m^T \otimes C_m^H)$ acts as the specific measurement matrix for that individual training instant [31].

By concatenating the received signals across all N_p successive training instants, the entire

observation model is consolidated into a single linear system:

$$\begin{bmatrix} y_1 \\ \vdots \\ y_m \\ \vdots \\ y_{N_p} \end{bmatrix} = \begin{bmatrix} \Phi_1 \\ \vdots \\ \Phi_m \\ \vdots \\ \Phi_{N_p} \end{bmatrix} h + \begin{bmatrix} C_1^H & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & C_{N_p}^H \end{bmatrix} \begin{bmatrix} n_1 \\ \vdots \\ n_m \\ \vdots \\ n_{N_p} \end{bmatrix} \quad (3.18)$$

To formalize this mathematically, The aggregate variables from the above system are defined as:

$$\tilde{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_{N_p} \end{bmatrix}, \quad \mathbf{\Phi} = \begin{bmatrix} \Phi_1 \\ \vdots \\ \Phi_{N_p} \end{bmatrix}, \quad \mathbf{\Psi} = \text{diag}(C_1^H, \dots, C_{N_p}^H), \quad \tilde{n} = \begin{bmatrix} n_1 \\ \vdots \\ n_{N_p} \end{bmatrix} \quad (3.19)$$

Substituting these definitions into the stacked representation yields the fundamental linear system utilized for channel estimation:

$$\tilde{y} = \mathbf{\Phi}h + \mathbf{\Psi}\tilde{n} \quad (3.20)$$

In this consolidated model, $\tilde{y} \in \mathbb{C}^{N_s N_p \times 1}$ denotes the aggregate received signal vector, $\mathbf{\Phi} \in \mathbb{C}^{N_s N_p \times N_t N_r}$ represents the comprehensive measurement matrix, $\mathbf{\Psi} \in \mathbb{C}^{N_s N_p \times N_p N_r}$ serves as the projection matrix for the noise, and $\tilde{n} \in \mathbb{C}^{N_p N_r \times 1}$ is the stacked noise vector [31]. Because the total number of pilot signals N_p is significantly smaller than the massive dimensionality of h , recovering the channel directly via mathematical inversion is ill-posed, requiring structural priors to solve.

3.4.2 Dictionary Constraints and Grid-Mismatch

Conventional CS approaches attempt to solve the underdetermined system formulated above by discretizing the continuous physical space into a fixed analytical dictionary grid. In the near-field regime, the array manifold matrices must account for both angular directions and absolute distances [31].

The spatial domain is divided into a discrete grid of size $G_r = G_{r,\phi} G_{r,\theta} G_{r,d}$ for the receiver and $G_t = G_{t,\phi} G_{t,\theta} G_{t,d}$ for the transmitter. The predefined grid points for the elevation (θ), azimuth

(ϕ) , and distance (d) are uniformly distributed such that:

$$\theta \in \left\{0, \frac{\pi}{2G_\theta}, \dots, \frac{\pi(G_\theta - 1)}{2G_\theta}\right\} \quad (3.21)$$

$$\phi \in \left\{0, \frac{2\pi}{G_\phi}, \dots, \frac{2\pi(G_\phi - 1)}{G_\phi}\right\} \quad (3.22)$$

$$d \in \left\{d_{\min}, d_{\min} + \frac{d_{\max} - d_{\min}}{G_d}, \dots, d_{\max} - \frac{d_{\max} - d_{\min}}{G_d}\right\} \quad (3.23)$$

where G_θ , G_ϕ , and G_d denote the grid resolution sizes for each respective dimension. Using all possible combinations of these discrete parameters, the overall near-field dictionary matrices $\bar{A}_R \in \mathbb{C}^{N_r \times G_r}$ and $\bar{A}_T \in \mathbb{C}^{N_t \times G_t}$ are constructed [31].

Using these dictionary matrices, the physical channel matrix can be approximated as a sparsely populated matrix $\bar{G}$ containing L non-zero elements:

$$H \approx \bar{A}_R \bar{G} \bar{A}_T^H \quad (3.24)$$

By applying the vectorization property, the vectorized physical channel is represented as $h \approx (\bar{A}_T^* \otimes \bar{A}_R) \bar{g} = \mathcal{D} \bar{g}$, where $\mathcal{D}$ is the massive global dictionary matrix and $\bar{g}$ is the sparse path gain vector. Substituting this into our observation model yields the standard CS optimization problem:

$$\tilde{y} = \Phi \mathcal{D} \bar{g} + \Psi \tilde{n} \quad (3.25)$$

Although this formulation theoretically allows for convex l_1 -norm minimization recovery, the predefined spatial grid introduces severe physical bottlenecks. The near-field spherical wave model adds distance as a third physical dimension alongside the azimuth and elevation angles, expanding the dictionary search space exponentially. Furthermore, true physical channel parameters continuously vary and rarely align perfectly with the discrete grid elements. This discrepancy forces the signal's energy to smear across multiple adjacent grid points, destroying the strict mathematical assumption of sparsity and introducing severe grid-mismatch errors that classical iterative algorithms cannot resolve [31].

3.5 Conventional CS Recovery Algorithms

To establish a rigorous mathematical baseline for near-field channel estimation, The recovery process using classical sparse recovery algorithms is formulated. Because the system $\tilde{y} = \Phi\mathcal{D}\bar{g} + \Psi\tilde{n}$ is severely underdetermined, standard matrix inversion is intractable. The problem is instead treated as an l_0 -norm minimization task:

$$\min \|\bar{g}\|_0 \quad \text{s.t.} \quad \|\tilde{y} - \Phi\mathcal{D}\bar{g}\|_2^2 \leq \epsilon \quad (3.26)$$

where ϵ is the noise tolerance. To solve this, greedy pursuit algorithms are conventionally employed to iteratively recover the sparse path gain vector [33].

3.5.1 Orthogonal Matching Pursuit (OMP) and SOMP

The fundamental classical approach for solving this near-field CS problem is the Orthogonal Matching Pursuit (OMP) algorithm. OMP iteratively projects the received signal $\tilde{y}$ onto the columns of the massive effective dictionary matrix $\Phi\mathcal{D}$. At each iteration, it selects the specific column most highly correlated with the current residual signal, sequentially building the support of the sparse vector [33].

However, applying standard OMP independently across each subcarrier neglects the physical properties of the wideband channel. To exploit the geometric reality that spatial paths (AoD, AoA, and distance) remain structurally consistent across frequencies, Simultaneous OMP (SOMP) is utilized. SOMP processes the received signals $y[k]$ from all K subcarriers jointly, enforcing a shared spatial support $\text{supp}(\bar{g}[k])$ across the entire bandwidth. This joint processing significantly improves estimation accuracy under severe noise conditions [33].

3.5.2 Dictionary Reduction (DR) Method

While SOMP improves multicarrier estimation, the sheer dimensionality of the near-field dictionary $\mathcal{D}$ creates an exhaustive computational burden. To mitigate this within a sub-connected AoSA architecture, a Dictionary Reduction (DR) method is employed by exploiting the high spatial correlation between adjacent subarrays [33]. The exact mathematical procedure operates as follows:

1. **Initial Subarray Estimation:** The SOMP algorithm is first executed exclusively for

the first Tx-Rx SA pair. Using the received signal $y^{1,1}[k]$ across all subcarriers and the full, massive dictionary grid, the algorithm isolates the initial sparse support, denoted as $\text{supp}(\bar{a}_{1,1})$.

2. **Oversampling and Truncation:** To refine the continuous angles, an oversampled dictionary $\bar{A}_{OVS}$ is mathematically constructed using an oversampling factor $G_{OVS} = G_{OVS}^h \times G_{OVS}^v$. Instead of searching the entire spatial domain, localized reduced dictionaries ($A_{T,DR}$ and $A_{R,DR}$) are generated by retaining only a highly limited number of columns (G) from $\bar{A}_{OVS}$. These columns strictly correspond to the fine spatial angles centered immediately around the initial AoD and AoA estimates derived from $\text{supp}(\bar{a}_{1,1})$ [33].
3. **Sequential SA Recovery:** For all remaining Rx SAs (indexed $2, \dots, Q_R$), the CS problem is solved using the drastically reduced Kronecker dictionary:

$$\bar{\Theta}_{DR} = (A_{T,DR} \otimes A_{R,DR}) \quad (3.27)$$

This constrained search space is subsequently repeated for all Q_T Tx SAs, massively reducing computational overhead.

3.5.3 Fundamental Limitations of Classical Frameworks

By constraining the search space, the SOMP-DR method achieves substantial improvements in NMSE compared to standard OMP [33]. However, despite these mathematical optimizations, analytical dictionary-based methods remain fundamentally bounded by the grid-mismatch problem. Since true physical AoDs, AoAs, and distances are continuous functions, they rarely align perfectly with the predefined discrete columns of the reduced dictionary $\bar{\Theta}_{DR}$. This spatial discrepancy forces the signal's energy to smear across multiple adjacent grid points, destroying the strict mathematical assumption of sparsity and introducing severe estimation error floors that classical iterative algorithms cannot circumvent.

3.6 Deep Compressed Sensing (DCS) Framework

To overcome the fundamental grid-mismatch limitations inherent in classical algorithms such as SOMP-DR [33], A transition from an analytical spatial matrix to a data-driven generative prior is implemented. The proposed DCS framework abandons the discrete dictionary $\mathcal{D}$ entirely. Instead, it leverages a GAN to learn the continuous, underlying physical distribution of the THz UM-MIMO channel.

3.6.1 The Continuous Generative Prior

In classical CS, the physical channel is mathematically forced into a sparse combination of predefined grid points, denoted as $h \approx \mathcal{D}\bar{g}$. In the proposed DCS architecture, this rigid constraint is replaced by a deep generative network:

$$h \approx G_{\vartheta}(z) \quad (3.28)$$

where $z \in \mathbb{R}^k$ represents a low-dimensional latent space vector, and $G_{\vartheta}(\cdot)$ is the Generator network parameterized by its trained weights ϑ . Because the neural network defines a continuous, non-linear mapping from the latent space directly to the high-dimensional channel space, the generated spatial paths are not forced to snap to an artificial grid, effectively eliminating the grid-mismatch error.

3.6.2 Offline Channel Distribution Learning

To ensure the Generator G_{ϑ} accurately models the physical properties of the near-field THz channel, it undergoes an adversarial training process against a Discriminator network D_{ω} . During the offline phase, the Generator maps random noise vectors $z \sim \mathcal{N}(0, I)$ into synthetic channel matrices, while the Discriminator attempts to distinguish between these and a dataset of mathematically simulated physical channels.

The networks are optimized iteratively using the standard min-max adversarial loss function:

$$\min_G \max_D V(D, G) = \mathbb{E}_{h \sim p_{data}(h)} [\log D_{\omega}(h)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D_{\omega}(G_{\vartheta}(z)))] \quad (3.29)$$

Once the offline training reaches convergence, the Discriminator is discarded, and the Generator's learned parameters ϑ are frozen to serve as the structural prior for the online phase.

3.6.3 Online Latent Space Optimization

During the real-time communication phase, the receiver must estimate the instantaneous channel h from the severely underdetermined pilot signal $\tilde{y}$. Because the Generator's weights ϑ are fixed, the classical l_0 -norm search for a massive sparse vector $\bar{g}$ is entirely bypassed. Instead, the problem shifts to optimizing the low-dimensional latent vector z :

$$z^* = \arg \min_z \|\tilde{y} - \Phi G_\vartheta(z)\|_2^2 + \lambda \|z\|_2^2 \quad (3.30)$$

where λ is a regularization parameter. Unlike the NP-hard l_0 -minimization problem, this generative objective function is continuously differentiable. Consequently, the optimal latent vector z^* can be calculated efficiently using standard gradient descent algorithms. The final estimated channel is recovered by passing the optimized latent vector through the generator: $\hat{h} = G_\vartheta(z^*)$.

3.7 Meta-Learning Enhanced DCS Architecture

While the base DCS framework provides a continuous generative prior to eliminate grid-mismatch errors, wireless environments are fundamentally dynamic. Traditional offline training captures a static data distribution, which can lead to slow convergence during online inference. To overcome this, A Model-Agnostic Meta-Learning (MAML) is integrated into the DCS framework. MAML explicitly trains the neural network to rapidly adapt and optimize its own inference process with minimal computational overhead.

3.7.1 Inner and Outer Loop Optimization

The MAML algorithm improves the speed of signal recovery by dividing the training into two distinct computational loops. During the **inner optimization**, the network takes a specific channel realization task and performs a fixed number of gradient descent steps on the latent vector z :

$$z \leftarrow z - \alpha_1 \nabla_z \mathcal{L}_\vartheta(\tilde{y}, \Phi, z) \quad (3.31)$$

where α_1 is the inner learning rate. Following each gradient step, the latent vector is normalized ($z \leftarrow z/\|z\|_2$) to maintain stability.

Following the inner loop, the **outer optimization** updates the actual parameters of the Generator network (ϑ) to minimize the expected NMSE across all tasks:

$$\min_{\vartheta} \mathcal{L}_G, \quad \text{where} \quad \mathcal{L}_G = \mathbb{E}_{h \sim \mathbb{P}}[\text{NMSE}(h, G_{\vartheta}(z^*))] \quad (3.32)$$

3.7.2 Restricted Isometry Property (RIP) Enforcement

To guarantee successful channel reconstruction, the measurement matrix must satisfy the Restricted Isometry Property (RIP), ensuring that the distance information between sparse signals is preserved upon projection. To strictly enforce this property during MAML training, an auxiliary measurement loss function is introduced, $\mathcal{L}_F$, evaluated over pairs of sampled channels (h_1, h_2):

$$\mathcal{L}_F = \mathbb{E}_{h_1, h_2} \left[\left(\frac{\|\Phi(h_1 - h_2)\|_2^2}{\|h_1 - h_2\|_2^2} - 1 \right)^2 \right] \quad (3.33)$$

By incorporating this RIP constraint, the final global update rule for the Generator's parameters becomes:

$$\vartheta \leftarrow \vartheta - \alpha_2 \nabla_{\vartheta} (\mathcal{L}_G + \mathcal{L}_F) \quad (3.34)$$

where α_2 is the outer learning rate. By explicitly optimizing for fast adaptation via MAML, the system eliminates the exhaustive combinatorial search times required by conventional CS.

3.8 Summary of the Proposed DCS-MAML Algorithm

To synthesize the mathematical formulations presented in the preceding sections, the complete Deep Compressed Sensing framework enhanced with Model-Agnostic Meta-Learning (DCS-MAML) is summarized below. The system is structurally divided into two primary operations: the offline meta-training phase and the online real-time inference phase.

The holistic system architecture of the proposed framework is illustrated in Figure 3.3. Unlike classical dictionary-based methods, the received pilots are routed directly through a continuous generative prior. During the offline phase, the Generator is trained adversarially to capture the physical channel distribution. Once deployed for online inference, the architecture rapidly optimizes the latent space to reconstruct the real-time channel.

To ensure the Generator adapts rapidly during the online phase without suffering from mode collapse or slow convergence, the offline training incorporates Model-Agnostic Meta-Learning.

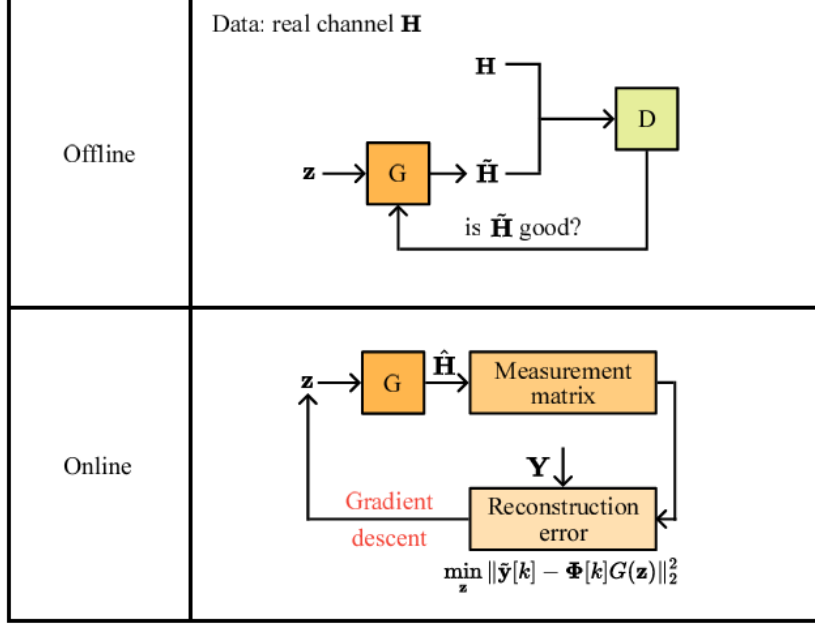


Figure 3.3: System architecture of the proposed DCS estimator [31].

Figure 3.4 details the specific MAML optimization loops governing this process. It highlights the task-specific inner loop updates (which optimize the latent vector over T gradient steps) and the global outer loop that updates the Generator's parameters to minimize the expected reconstruction error.

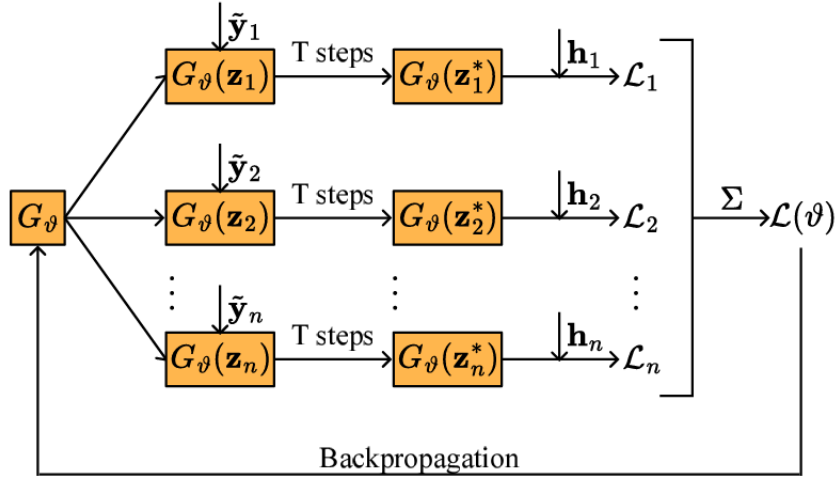


Figure 3.4: Block diagram of the offline MAML training process for the DCS framework [31].

3.8.1 Algorithmic Implementation

The exact pseudo-code for the offline meta-training process is detailed in Algorithm 1. This phase is computationally intensive but is performed exclusively offline at the base station or a

centralized server prior to deployment.

Algorithm 1 Offline Meta-Training of the DCS Generator

Require: Dataset of true channel matrices $\mathcal{H}_{train}$, Measurement matrix Φ , Inner learning rate α_1 , Outer learning rate α_2 , Number of inner steps N_{inner} .

Ensure: Optimal Generator weights ϑ^* .

- 1: Initialize Generator G_{ϑ} with random weights.
 - 2: **while** not converged **do**
 - 3: Sample a batch of channel realization tasks $\{h_i\}_{i=1}^B$ from $\mathcal{H}_{train}$.
 - 4: **for** each task h_i **do**
 - 5: Simulate received pilot signal: $\tilde{y}_i = \Phi h_i + \tilde{n}$.
 - 6: Initialize latent vector $z_i \sim \mathcal{N}(0, I)$.
 - 7: **for** $j = 1$ to N_{inner} **do**
 - 8: Compute inner loss: $\mathcal{L}_{inner} = \|\tilde{y}_i - \Phi G_{\vartheta}(z_i)\|_2^2 + \lambda \|z_i\|_2^2$
 - 9: Gradient update: $z_i \leftarrow z_i - \alpha_1 \nabla_{z_i} \mathcal{L}_{inner}$
 - 10: Normalize latent vector: $z_i \leftarrow z_i / \|z_i\|_2$
 - 11: **end for**
 - 12: Save the optimized latent vector as z_i^* .
 - 13: **end for**
 - 14: Compute base reconstruction loss: $\mathcal{L}_G = \frac{1}{B} \sum_{i=1}^B \text{NMSE}(h_i, G_{\vartheta}(z_i^*))$
 - 15: Compute RIP auxiliary loss $\mathcal{L}_F$ to enforce spatial constraints.
 - 16: Outer meta-update: $\vartheta \leftarrow \vartheta - \alpha_2 \nabla_{\vartheta} (\mathcal{L}_G + \mathcal{L}_F)$
 - 17: **end while**
 - 18: **Return** trained generator weights ϑ^* .
-

Once the generator weights ϑ^* are optimized and frozen, the highly efficient online inference phase is deployed in the field. Because the network has explicitly learned how to optimize itself via MAML, the latent vector z converges rapidly, as detailed in Algorithm 2.

Algorithm 2 Online Real-Time Channel Estimation (Inference Phase)

Require: Frozen Generator G_{ϑ^*} , Real-time received pilot signal $\tilde{y}$, Measurement matrix Φ , Learning rate α_{online} , Maximum iterations K_{max} .

Ensure: Estimated channel vector $\hat{h}$.

- 1: Initialize latent vector $z \sim \mathcal{N}(0, I)$.
 - 2: **for** $k = 1$ to K_{max} **do**
 - 3: Compute gradient: $g = \nabla_z (\|\tilde{y} - \Phi G_{\vartheta^*}(z)\|_2^2 + \lambda \|z\|_2^2)$
 - 4: Update latent vector: $z \leftarrow z - \alpha_{online} \cdot g$
 - 5: Project onto unit sphere: $z \leftarrow z / \|z\|_2$
 - 6: **if** $\|\nabla_z\|_2 < \epsilon$ **then**
 - 7: **break**
 - 8: **end if**
 - 9: **end for**
 - 10: Generate final channel estimate: $\hat{h} = G_{\vartheta^*}(z)$
 - 11: **Return** $\hat{h}$.
-

Chapter 4

Performance Evaluation

This chapter will highlight the effectiveness of the proposed DCS framework, by utilizing various metrics to compare it to conventional methods and illustrating results of simulations.

4.1 Methodology Overview

To establish a clear foundation for this analysis, the overall system workflow is presented in Figure 4.1. Following the initial system setup which defines the THz UM-MIMO near-field channel and observation model the methodology branches to compare two distinct channel estimation frameworks:

- **Conventional CS (Baseline):** A traditional approach relying on spatial dictionaries and iterative greedy pursuit to extract the channel estimate based on a predefined sparsity level.
- **Deep Compressed Sensing (Proposed):** The DCS-MAML framework, which leverages offline meta-training to optimize generator weights, followed by an online inference phase that utilizes gradient descent to refine the latent vector until a specific error tolerance is met.

Finally, the outputs from both frameworks are benchmarked in the performance evaluation stage to assess their relative accuracy, reliability, and computational efficiency.

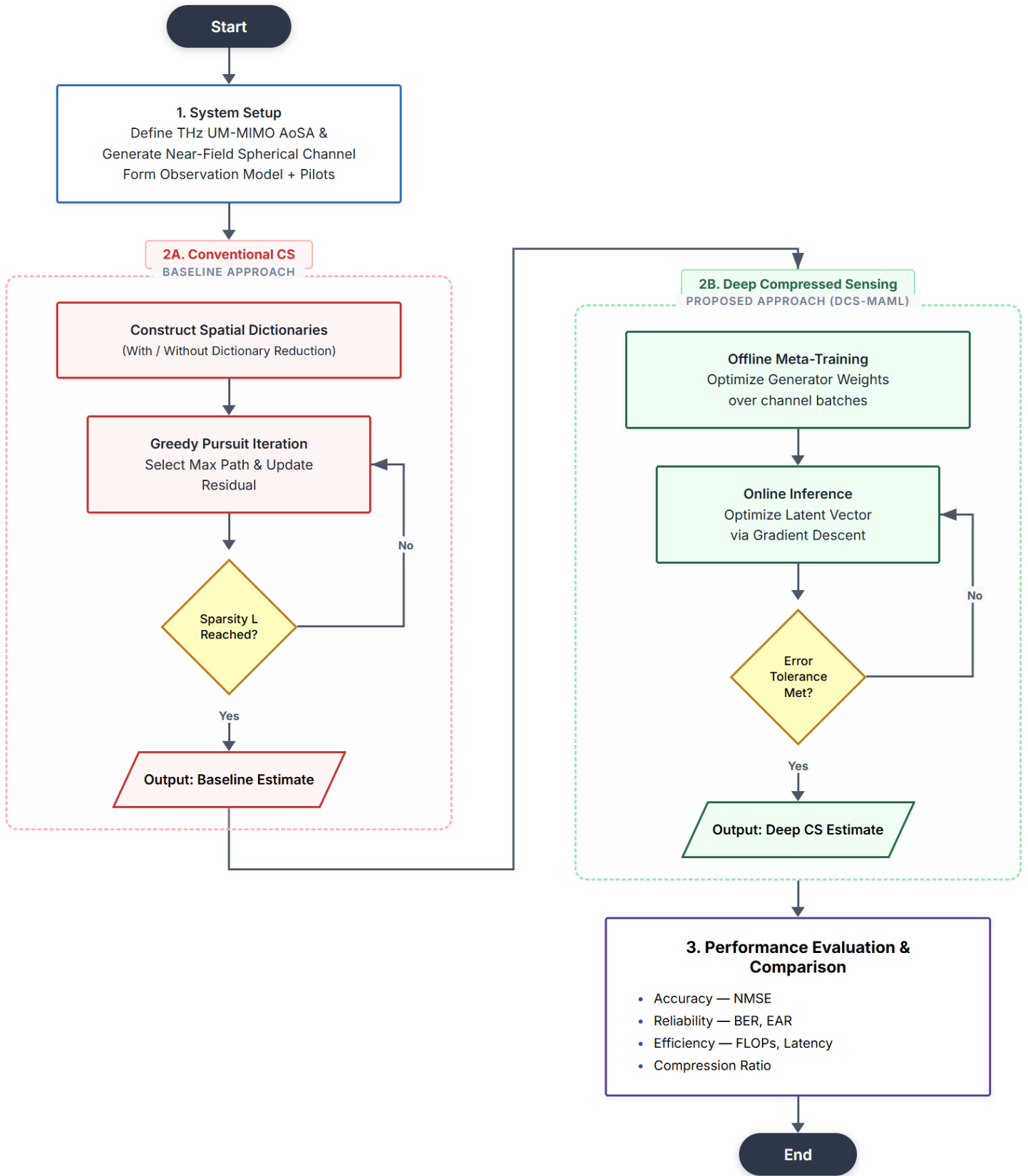


Figure 4.1: Overall methodology flowchart

4.1.1 Methodology Flowchart Overview

The flowchart outlined in Figure 4.1 illustrates the step-by-step methodology used to evaluate and compare channel estimation frameworks. The workflow is structured into three primary phases:

1. System Setup

The process begins by establishing the physical and mathematical foundations for the simulation. In this initial step, the Terahertz (THz) Ultra-Massive MIMO system is defined utilizing an Array-of-Subarrays (AoSA) configuration. Once the physical architecture is set, the system generates the near-field spherical channel models and constructs the linear observation model alongside the pilot signals required for training the estimators.

2. Parallel Estimation Approaches

Following the setup, the methodology branches into two distinct paths to benchmark a traditional mathematical framework against the proposed deep learning-based model.

- **2A. Conventional CS (Baseline Approach):** This path represents a traditional Compressed Sensing method. The system constructs spatial dictionaries (with or without dictionary reduction) and enters an iterative “Greedy Pursuit” loop. This loop selects the maximum path and updates the residual error until a predefined mathematical sparsity level is reached, producing the baseline estimate.
- **2B. Deep Compressed Sensing (Proposed Approach):** This path represents the proposed DCS-MAML framework. It begins with an “Offline Meta-Training” phase to optimize the generator’s weights across channel batches. Once trained, it moves to “Online Inference,” where gradient descent is used to refine a latent vector until a specific error tolerance is met, yielding the Deep CS estimate.

3. Performance Evaluation & Comparison

In the final phase, the channel estimates generated by both frameworks are benchmarked against one another. The system evaluates the performance across three main categories:

- **Accuracy:** Evaluated using the Normalized Mean Square Error (NMSE).

- **Reliability:** Assessed through the Bit Error Rate (BER) and Effective Achievable Rate (EAR).
- **Efficiency:** Measured by tracking computational complexity via Floating-Point Operations (FLOPs), actual execution latency, and the Compression Ratio.

4.2 Evaluation Metrics

To evaluate the effectiveness of the proposed DCS framework, this section outlines the key performance metrics used in the analysis. These metrics provide quantitative insight into trained model, computational cost, and overall communication quality.

4.2.1 Normalized Mean Squared Error

The accuracy of the channel estimation is quantified using the Normalized Mean Square Error (NMSE), which evaluates the average squared error between the true channel $\mathbf{h}[k]$ and the estimated channel $\hat{\mathbf{h}}[k]$ across all K subcarriers. The NMSE is mathematically defined as:

$$\text{NMSE} = \frac{\sum_{k=0}^{K-1} \|\mathbf{h}[k] - \hat{\mathbf{h}}[k]\|_2^2}{\sum_{k=0}^{K-1} \|\mathbf{h}[k]\|_2^2} \quad (4.1)$$

4.2.2 Bit Error Rate (BER)

To evaluate the communication reliability of the proposed framework, the Bit Error Rate (BER) is used as a key performance metric. BER measures the ratio of incorrectly detected bits to the total number of transmitted bits, and is defined as

$$\text{BER} = \frac{N_{\text{err}}}{N_{\text{bits}}}, \quad (4.2)$$

where N_{err} is the number of erroneous bits after detection and demodulation, and N_{bits} is the total number of transmitted bits.

In this work, BER is evaluated through Monte Carlo simulations by transmitting modulated symbols over the channel and performing signal detection using the estimated channel information. Therefore, BER provides a practical end-to-end measure of how channel estimation accuracy affects link reliability.

4.2.3 Effective Achievable Rate (EAR)

The effective achievable rate (EAR) is used to evaluate the practical throughput of the communication system. It measures how much information can be reliably transmitted over the channel while also accounting for the loss in transmission resources caused by pilot-based channel training. Therefore, EAR is a more realistic performance metric than an ideal rate measure, since it reflects not only the channel quality but also the overhead required to estimate the channel.

In this work, the EAR is defined as

$$R_{\text{eff}} = \frac{\eta}{EK} \sum_{e=1}^E \sum_{k=1}^K \log_2 \left| \mathbf{I}_{Q_R} + \frac{\bar{P}_T}{KN_S} \mathbf{R}_n^{-1} \mathbf{C}^H[k] \mathbf{C}[k] \right|, \quad (4.3)$$

where E is the number of independent simulation trials, K is the number of subcarriers, Q_R is the number of receive subarrays, $\bar{P}_T$ is the average transmit power during data transmission, and N_S is the number of transmitted data streams. The matrix $\mathbf{I}_{Q_R}$ denotes the identity matrix of size Q_R .

The equivalent channel term is expressed as

$$\mathbf{C}[k] = \hat{\mathbf{F}}^H[k] \mathbf{H}^H[k] \hat{\mathbf{W}}[k], \quad (4.4)$$

where

$$\hat{\mathbf{F}}[k] = \hat{\mathbf{F}}_{\text{RF}} \hat{\mathbf{F}}_{\text{BB}}[k], \quad (4.5)$$

and $\hat{\mathbf{W}}[k]$ is defined in a similar manner. The noise covariance matrix is given by

$$\mathbf{R}_n = \sigma_n^2 \hat{\mathbf{W}}^H[k] \hat{\mathbf{W}}[k], \quad (4.6)$$

where σ_n^2 is the noise variance.

The factor η represents the efficiency of data transmission after training and is defined as

$$\eta = \max \left(0, 1 - \frac{M_T^{\text{tr}}}{T_{\text{coh}}} \right), \quad (4.7)$$

where M_T^{tr} is the number of transmit pilot symbols used during training, and T_{coh} is the channel coherence time in symbols. This factor reduces the rate according to the portion of the

coherence interval consumed by training, since only the remaining symbols are available for data transmission.

Hence, the EAR represents the effective SE that can be achieved in practice after considering both channel estimation and training overhead. A higher value of R_{eff} indicates that the system can support more efficient and reliable data transmission under the available channel knowledge.

4.2.4 Compression Ratio

The compression ratio, denoted as ρ , is a fundamental metric used to evaluate the resource efficiency of the pilot training phase during channel estimation. In UM-MIMO systems, acquiring full CSI using traditional orthogonal pilot sequences requires an overhead that scales with the number of antennas. This severely bottlenecks practical throughput. The compression ratio measures the degree to which this training overhead can be reduced by exploiting the inherent structural properties of the channel, such as spatial sparsity.

In this work, the compression ratio is defined as

$$\rho = \frac{N_p}{N_t}, \quad (4.8)$$

where N_p is the number of transmitted pilot symbols (or measurements) utilized during the training phase, and N_t is the total number of transmit antennas at the base station.

A lower ρ indicates a shorter training duration, which directly increases the efficiency factor η defined in Equation (4.7). By consuming less of the coherence interval T_{coh} for training, a larger portion of the communication block is reserved for actual data transmission.

However, reducing N_p means the estimator has fewer observations to reconstruct the high-dimensional channel. For conventional compressive sensing algorithms, such as OMP and SOMP, decreasing the compression ratio leads to a rapid degradation in estimation accuracy (i.e., a higher NMSE).

Therefore, a highly effective channel estimation scheme such as DCS is characterized by its ability to maintain a robust, low NMSE even at low compression ratios. Achieving high accuracy at a low compression ratio ensures that the system minimizes pilot overhead without sacrificing the reliability of the reconstructed equivalent channel $C[k]$.

4.2.5 Computational Complexity

Computational complexity is a critical evaluation metric used to assess the practical deployment viability of the channel estimation algorithms in real-time THz networks. UM-MIMO systems process an immense amount of spatial data, requiring the estimation algorithm to execute rapidly to prevent critical latency delays and ensure the channel is estimated well within the coherence time.

In this work, computational complexity is evaluated using two primary approaches:

Theoretical Complexity (*O*-notation) and FLOPs

This dimension quantifies the mathematical burden of the algorithms by calculating the required number of Floating-Point Operations (FLOPs). To evaluate this theoretically, Big-*O* notation is employed. Big-*O* notation describes the asymptotic upper bound of an algorithm's computational load, isolating the dominant variables that dictate how the runtime scales as the system dimensions such as the number of antennas, grid resolution, or sparsity level grow toward infinity. By discarding constant coefficients and lower-order terms, it provides a fundamental mathematical framework for comparing how efficiently different algorithms handle massive datasets.

For classical dictionary-based methods, the complexity is dominated by the massive, exhaustive correlation steps between the predefined spatial dictionaries of size G and the residual signals. For example, the complexity of standard OMP scales as:

$$O(K(LN_pG + L^3)) \quad (4.9)$$

where K is the number of subcarriers, L is the sparsity level, and N_p is the number of pilot symbols.

In contrast, the proposed Deep Compressed Sensing (DCS) framework bypasses this exhaustive grid search by shifting the heavy mathematical burden entirely to an offline meta-learning phase. During real-time online inference, the computational complexity is strictly a function of the number of gradient descent steps T required to optimize the latent vector, utilizing highly parallelizable matrix multiplications rather than combinatorial searches. This yields

a drastically reduced online theoretical complexity of:

$$O(TN_sN_pN_tN_r) \quad (4.10)$$

where N_s is the number of data streams, and N_t and N_r represent the transmit and receive antennas.

Average Online Runtime

While FLOPs and Big- O notation provide a theoretical bound, the average online runtime translates this mathematical complexity into physical execution latency measured in seconds. This metric provides a direct, system-level evaluation of whether the estimator can deliver accurate Channel State Information (CSI) swiftly enough to support time-sensitive applications. A highly effective estimation scheme minimizes this execution time, bridging the massive latency gap inherent in unoptimized conventional search algorithms and ensuring continuous, real-time deployment.

4.3 Dataset Preparation and Normalization

A wideband system is considered where the main simulation parameters are summarized in Table 4.1. In our simulation configuration, the Tx and Rx antenna apertures are defined as $D_t = 14\lambda$ and $D_r = 12\lambda$, respectively. Consequently, the near-field Rayleigh distance threshold is determined by:

$$d_{\text{Rayleigh}} = \frac{2(D_t + D_r)^2}{\lambda} = 1.35 \text{ m}. \quad (4.11)$$

To ensure the presence of near-field effects, the propagation distance between the Tx and Rx is set to 1 m. While the original TeraMIMO simulator employs a spherical wave model at the SA level and a planar wave model at the AE-level to reduce computational complexity, the framework is modified to implement an AE-level spherical wave model for enhanced modeling accuracy.

In the default simulator settings, the Tx and Rx antenna arrays are parallel; however, to simulate more realistic and complex wireless environments, Rx array rotation is introduced across all three axes with a maximum permitted angle of $\phi = \pi/6$ in each dimension. Molecular absorption is also integrated into the dataset, utilizing the High-Resolution Transmission

Molecular Absorption (HITRAN) database.

Table 4.1: Main system parameters

Parameter	Value
Carrier frequency	$f_c = 300$ GHz
Bandwidth	$B = 10$ GHz
Number of subcarriers	$K = 8$
Number of Tx/Rx RF chains	$N_{\text{RF}}^{(t)} = 4, N_{\text{RF}}^{(r)} = 4$
Number of Tx/Rx AEs per SA	$N_{\text{AE}}^{(t)} = 64, N_{\text{AE}}^{(r)} = 8$
Number of Tx/Rx antennas	$N_t = 256, N_r = 32$
Number of data streams	$N_s = 4$
AE spacing	$d_{\text{AE}} = \lambda_c/2 = 0.0005$ m
SA spacing	$d_{\text{SA}} = 10\lambda_c = 0.01$ m
Pilot length	$N_p = 100$
Compression ratio	$\rho = N_p/N_t = 39\%$
LoS path length	$d = 1$ m
Wave model	Spherical wave
Training dataset size	5000
Testing dataset size	1000

Furthermore, unlike existing studies that assume a fixed number of Non-Line-of-Sight (NLoS) paths, our dataset utilizes independent Poisson processes to model the arrival rates of clusters and rays. Specifically, a cluster arrival rate of $\Lambda = 0.13 \text{ ns}^{-1}$ and a ray arrival rate of $\lambda = 0.37 \text{ ns}^{-1}$ are adopted within a time margin of $T_m = 50$ ns. Under these parameters, the number of NLoS paths L follows a Poisson distribution $L \sim \text{Pois}(\lambda T_m)$, resulting in an expected value and variance of $E[L] = \lambda T_m = 18.5$. Since this value is significantly lower than the number of antennas $\min\{N_t, N_r\} = 64$, the considered THz channels remain statistically sparse in the angular domain.

TeraMIMO is utilized, a MATLAB-based THz channel simulator, with slight modification, to generate the channel dataset in the frequency domain, where the channel realization is of size (N, K, N_r, N_t) . Noted that DL algorithms work only on real samples, the real and imaginary parts are separated and stacked over the second dimension, i.e., the size of the real-valued channel is $(N, 2K, N_r, N_t)$, which is then used in PyTorch. The channel dataset is similar to a typical image dataset, with the second dimension being RGB color channels and the last two dimensions being spatial. Before feeding the data into the training process, channel-wise

normalization is performed, as normally used in image processing. More precisely, for a given dataset H of size N , the c -th channel of the dataset will be normalized by using

$$\mathbf{H}[:, c, :, :] \leftarrow \frac{\mathbf{H}[:, c, :, :] - \mu_c}{\sigma_c} \quad (4.12)$$

where the mean (μ_c) and standard deviation (σ_c) for each channel are computed across the entire dataset of N realizations as:

$$\mu_c = \frac{1}{NN_rN_t} \sum_{n=1}^N \sum_{n_r=1}^{N_r} \sum_{n_t=1}^{N_t} \mathbf{H}[n, c, n_r, n_t] \quad (4.13)$$

$$\sigma_c = \sqrt{\frac{1}{NN_rN_t - 1} \sum_{n=1}^N \sum_{n_r=1}^{N_r} \sum_{n_t=1}^{N_t} |\mathbf{H}[n, c, n_r, n_t] - \mu_c|^2} \quad (4.14)$$

The measurement matrix Φ is constructed by adopting a zero-mean circular symmetric Gaussian distribution for the pilot symbols. The digital precoder $\mathbf{F}_{BB}$ and combiner $\mathbf{C}_{BB}$ are initialized as identity matrices. Furthermore, to reflect practical hardware constraints, the phase shifters at both the transmitter and receiver are modeled with a one-bit quantization resolution. The optimized hyperparameters utilized in our experimental framework are summarized in Table 4.2. The Adam optimizer is employed for all stochastic gradient descent processes. Specifically, the model utilizes a learning rate of 0.1 for latent space optimization to ensure rapid convergence, while a more conservative learning rate of 0.0002 is applied to the model update phase to maintain training stability [31].

Table 4.2: Deep Compressed Sensing (DCS) Hyperparameters

Hyperparameter	Value
Latent optimization (Online Phase)	
Optimizer	Adam
Learning rate	0.1
Momentum	(0.9, 0.99)
Number of restarts	1
Model optimization (Offline Phase)	
Optimizer	Adam
Learning rate	0.0002
Momentum	(0.9, 0.99)
Batch size	50
Number of epochs	300

4.4 Simulation Results and Discussion

This section presents a comprehensive analysis of the simulation results to rigorously evaluate the performance of the proposed DCS framework. The primary objective is to demonstrate the superiority of the generative prior by benchmarking its channel estimation accuracy against a suite of conventional dictionary-based CS recovery algorithms. These baselines include standard Orthogonal Matching Pursuit (OMP), Simultaneous OMP (SOMP), and their Dictionary Reduction enhanced variants (OMP-DR and SOMP-DR).

4.4.1 Normalized Mean Square Error Versus SNR

Figure 4.2 illustrates the Normalized Mean Square Error (NMSE) of the channel estimation as a function of the Signal-to-Noise Ratio (SNR), evaluated over a wide range from -20 dB to 20 dB. This metric directly assesses the robustness of the proposed DCS architecture against its conventional analytical counterparts under varying noise conditions.

As expected, the channel estimation accuracy for all evaluated schemes improves (NMSE decreases) as the transmit SNR increases from -20 dB to 0 dB. However, a key observation from the figure is the profound performance gap between the data-driven framework and the classical analytical methods. The proposed Deep CS architecture consistently achieves the lowest NMSE across the entire SNR regime.

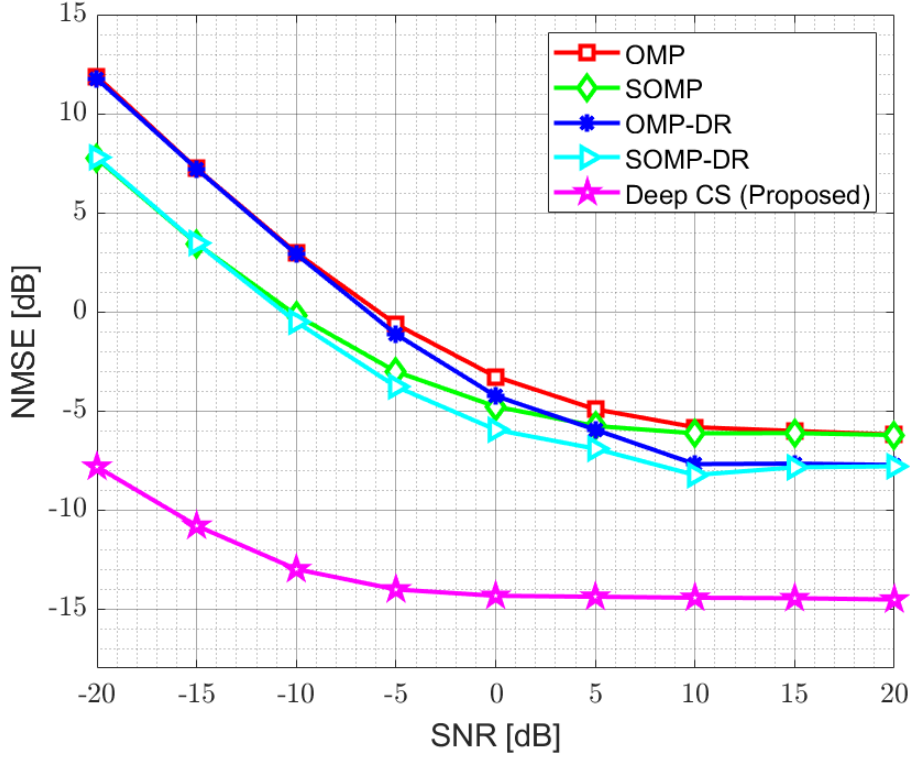


Figure 4.2: NMSE versus SNR for THz UM-MIMO channel estimation. The proposed DCS framework against OMP, SOMP, OMP-DR, SOMP-DR.

In the severe low-SNR regime (e.g., at -20 dB), the conventional algorithms experience catastrophic performance degradation, yielding NMSE values well above 5 dB. This indicates a near-complete failure to recover the true spatial paths from the noise-corrupted pilot signals. In contrast, the proposed Deep CS method maintains a highly robust NMSE of approximately -8 dB at the same noise level. This demonstrates that the pre-trained generative prior possess a deep structural understanding of the physical channel distribution, allowing it to effectively denoise and recover spatial features even under extreme interference.

Furthermore, the fundamental structural limitations of conventional methods become glaringly evident in the high-SNR regime (0 dB to 20 dB). While the SOMP-DR method slightly outperforms standard OMP by exploiting joint subcarrier sparsity and localized dictionaries, all conventional CS curves eventually flatten out, hitting an unavoidable "error floor." Specifically, OMP and SOMP saturate near an NMSE of -6 dB, while the DR-enhanced methods saturate near -8 dB. This error floor is the direct mathematical consequence of the grid-mismatch problem detailed in Chapter 3. Because the physical angles and distances of the near-field THz paths are continuous, they cannot perfectly align with the discrete points of the predefined

spatial dictionary, causing unresolvable quantization errors regardless of how high the signal quality becomes.

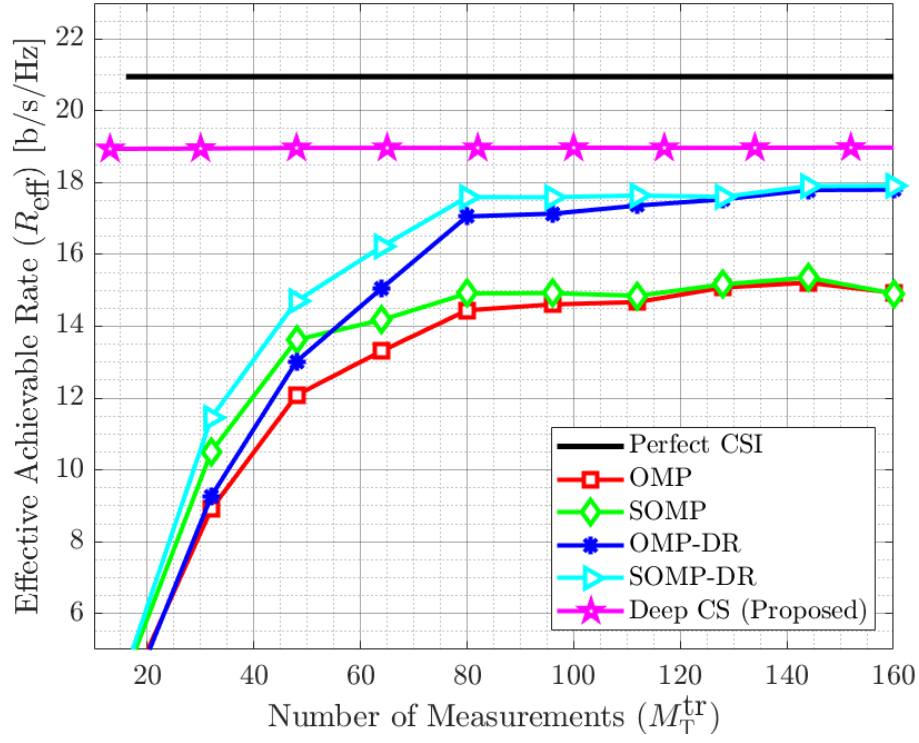
Conversely, the proposed Deep CS framework completely bypasses this limitation. By mapping the latent vector z directly to a continuous high-dimensional channel space, it is not bounded by an artificial grid. Consequently, the Deep CS curve does not suffer from the severe early saturation seen in the baselines, steadily pushing the estimation error floor down to nearly -15 dB at an SNR of 20 dB. This massive improvement validates that utilizing a continuous generative prior, optimized via meta-learning, is fundamentally superior to classical sparse recovery for near-field THz channel estimation.

4.4.2 Effective Achievable Rate Versus Number of Measurements

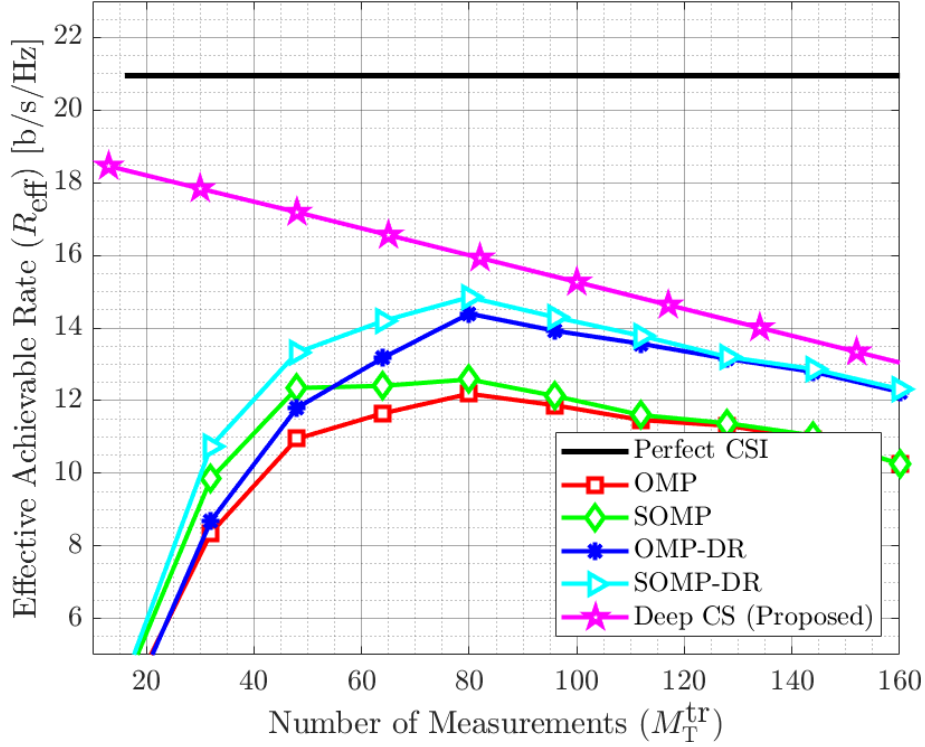
While NMSE isolates the mathematical accuracy of the channel estimation, the Effective Achievable Rate provides a holistic, system-level evaluation. Specifically, the factor η captures the loss in achievable rate due to the training phase. This creates a fundamental trade-off between channel estimation accuracy and training overhead; utilizing more pilot measurements improves the channel estimate but leaves fewer symbol slots available for actual data transmission within a given coherence time.

Figure 4.3 evaluates this trade-off by plotting the Effective Achievable Rate (R_{eff}) against the number of pilot measurements (M_7^{p}). To isolate the effects of estimation accuracy from the training overhead penalty, the system is evaluated under two distinct coherence time assumptions.

Figure 4.3a presents the theoretical scenario where the channel coherence time is assumed to be infinite ($T_{\text{coh}} = \infty$). In this regime, the overhead penalty factor η effectively equals 1, isolating how pure estimation accuracy limits the throughput. As established in the baseline literature, the DR-enhanced algorithms (SOMP-DR and OMP-DR) achieve a higher EAR than standard OMP and SOMP. However, all conventional CS algorithms perform poorly at low measurement regimes, requiring over 100 pilot symbols to asymptotically approach their maximum achievable rates. Conversely, the proposed Deep CS framework demonstrates an improved ability to reconstruct the channel from a severely underdetermined observation. It achieves a near-optimal, flat SE of approximately 19 b/s/Hz utilizing as few as 16 measurements, massively outperforming the conventional algorithms across the entire domain.



(a) $T_{\text{coh}} = \infty$



(b) $T_{\text{coh}} = 512$

Figure 4.3: Effective Achievable Rate versus the number of pilot measurements (M_T^{tr}). (a) Theoretical performance with infinite coherence time. (b) Realistic performance with a finite coherence block of 512 symbols.

Figure 4.3b translates this advantage into a highly realistic THz communication scenario featuring a rapid fading environment with a finite coherence block of $T_{\text{coh}} = 512$ symbols. Here, the fundamental pilot overhead trade-off dictates the curve geometries.

The conventional baseline methods exhibit a distinct concave shape, revealing an optimal training length. At extremely low measurement counts, their channel estimation is too poor to support high data rates. As M_T^{tr} increases, their estimation improves, raising the EAR. However, beyond an optimal "sweet spot," the fractional improvement in CSI accuracy is heavily outweighed by the severe loss of data transmission time. For the conventional DR methods, it has been shown that utilizing approximately 33% of the measurements compared to the system size is sufficient to achieve this trade-off [35]. This aligns perfectly with Figure 4.3b, where the SOMP-DR method peaks at roughly 80 measurements. Even at its absolute peak, the best conventional method only achieves an EAR of roughly 15 b/s/Hz.

In contrast, the proposed Deep CS architecture fundamentally breaks this compromise. Because the pre-trained generative network requires remarkably few pilots to accurately recover the channel, it reaches its peak SE immediately at the minimum evaluated measurement count ($M_T^{\text{tr}} \approx 16$). From this point onward, supplying the network with additional measurements does not meaningfully improve the already-excellent CSI, but simply incurs the mathematical overhead penalty η , resulting in the linear degradation observed in the plot. Consequently, the Deep CS framework allows the system to dedicate the vast majority of the 512-symbol coherence block strictly to payload data, achieving a peak EAR of over 18 b/s/Hz and demonstrating overwhelming superiority in environments where coherence times are exceptionally short.

4.4.3 Normalized Mean Square Error Versus Compression Ratio

While the previous analyses demonstrated robustness against noise, it is equally critical to evaluate the algorithms' performance with respect to the training overhead. The compression ratio, defined as $\rho = N_p/N_t$ (where N_p is the number of pilot measurements and N_t is the number of transmit antennas), dictates the efficiency of the channel estimator. A lower compression ratio implies fewer pilot symbols are required, which directly translates to reduced training overhead and higher effective data throughput.

Figure 4.4 illustrates the NMSE of the channel estimation as a function of the compression ratio ρ . This evaluates how much data each algorithm inherently needs to successfully reconstruct the high-dimensional THz channel.

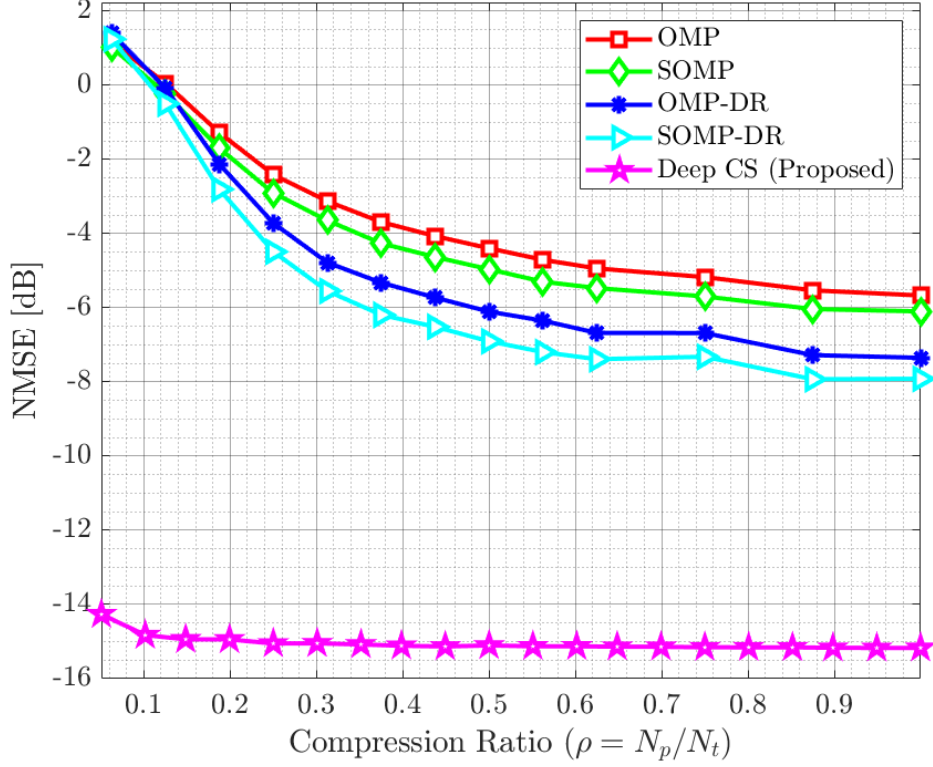


Figure 4.4: NMSE versus Compression Ratio ($\rho = N_p/N_t$).

The conventional CS methods (OMP, SOMP, and their DR variants) exhibit a steep, asymptotic learning curve. At highly compressed regimes (e.g., $\rho < 0.2$), these analytical methods suffer from severe performance degradation. Because they rely on classical sparse recovery, they are strictly bounded by fundamental sampling limits; they require a critical mass of measurements to accurately map the received signals onto their discrete spatial dictionaries. Even as the compression ratio increases towards 1.0, the conventional methods plateau early, saturating between -6 dB and -8 dB due to the previously established grid-mismatch errors.

In contrast, the proposed Deep CS framework exhibits superior pilot efficiency, fundamentally breaking the traditional sampling constraints. At the most extreme compression ratios (e.g., $\rho \approx 0.05$), the DCS network exhibits a slight initial error, as the gradient descent requires at least a minimal set of pilot projections to effectively optimize the latent vector z^* . However, because the generative prior already understands the physical channel manifold from its offline MAML training, it rapidly descends to its minimum error floor.

By $\rho \approx 0.15$, the Deep CS curve flattens out entirely, maintaining a highly accurate NMSE of approximately -15 dB across the remainder of the domain. This flatline indicates that once a sufficient baseline of pilots is provided to the generative model, supplying additional measurements yields no further improvement in estimation accuracy. By successfully recovering

the channel at such an exceptionally low compression threshold, the proposed DCS framework drastically minimizes the required communication overhead, leaving the vast majority of the coherence block available for payload data transmission.

4.4.4 Bit Error Rate Versus SNR and Spatial Multiplexing Limits

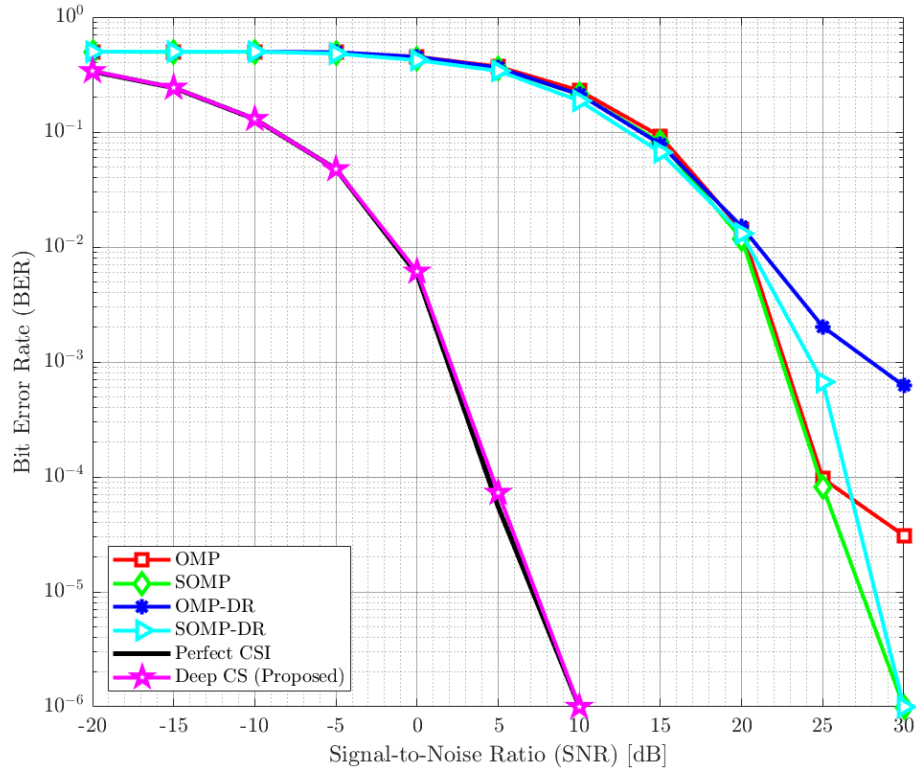
While NMSE and EAR evaluate the accuracy and theoretical capacity of the channel estimation, the Bit Error Rate (BER) provides the ultimate end-to-end measure of communication reliability. The accuracy of the estimated CSI directly dictates the performance of the hybrid precoders and combiners; poor CSI leads to beam misalignment and severe residual interference during symbol detection.

Figure 4.5 evaluates the BER performance of the proposed DCS framework against conventional dictionary-based algorithms under varying SNR conditions, employing 16-Quadrature Amplitude Modulation (16-QAM) for the transmitted data streams. To thoroughly investigate the system’s spatial multiplexing capabilities, the performance is analyzed for both a single data stream ($N_S = 1$) and multiple data streams ($N_S = 2$).

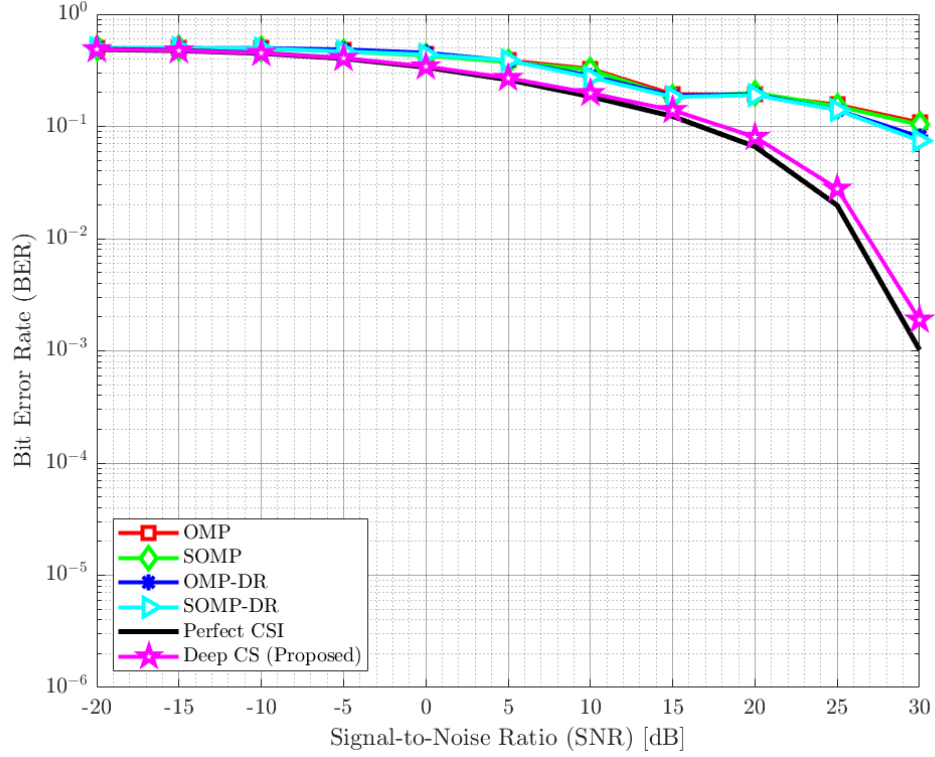
Figure 4.5a illustrates the system performance when transmitting a single data stream ($N_S = 1$). In this configuration, the entire UM-MIMO array is leveraged to maximize the beamforming gain toward a single spatial direction. As shown, the proposed Deep CS architecture significantly outperforms the conventional methods. By leveraging its highly accurate, continuous generative prior, the Deep CS framework achieves a target BER of 10^{-4} at an SNR of approximately 7 dB. In contrast, the conventional algorithms (OMP, SOMP, and their DR variants) require substantially higher SNR to reach similar reliability, with standard OMP failing to break the 10^{-3} barrier even at 20 dB due to grid-mismatch induced beam misalignment.

Figure 4.5b investigates the system’s performance when attempting to spatially multiplex two independent data streams ($N_S = 2$). An immediate observation is the overall performance degradation across all algorithms compared to the single-stream scenario. The Deep CS framework still maintains a distinct superiority, reaching a BER of 10^{-3} near 15 dB, while the conventional methods struggle to drop below 10^{-2} .

This overarching performance degradation is not a failure of the estimation algorithms, but rather a harsh coupling of fundamental physical constraints and strict architectural limitations. First, from a physical perspective, sub-millimeter THz channels are extremely sparse, heavily



(a) $N_S = 1$ Data Stream



(b) $N_S = 2$ Data Streams

Figure 4.5: BER versus SNR. (a) Single-stream transmission. (b) Multi-stream transmission.

dominated by the LoS path with very few viable NLoS clusters [41]. Consequently, the spatial degree-of-freedom is remarkably poor, resulting in a physical channel matrix ($\mathbf{H}$) with an inherently low mathematical rank.

Second, this environmental low-rank bottleneck is severely exacerbated by the hardware structure of the transceiver. To mitigate the immense power consumption and hardware cost of ultra-massive arrays, the system employs a sub-connected AoSA hybrid beamforming architecture [34, 40]. In the AoSA configuration, each RF chain is exclusively connected to a disjoint, localized subset of antennas. Mathematically, this imposes a strict block-diagonal constraint on the analog precoding and combining matrices.

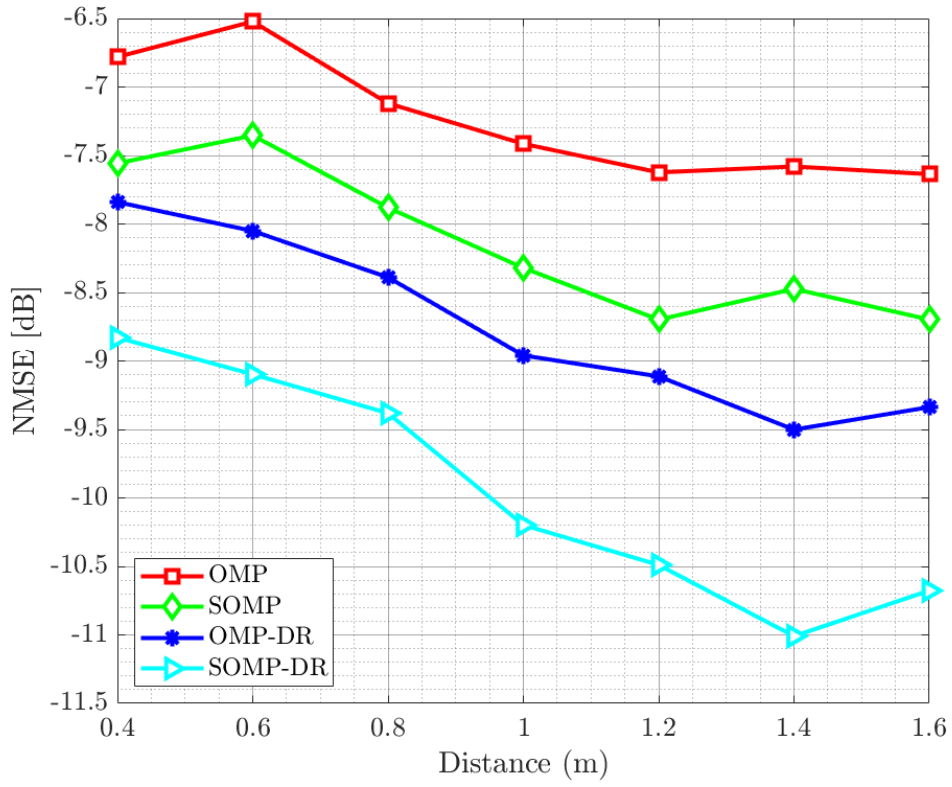
When the system attempts to transmit multiple data streams ($N_S \geq 2$), this block-diagonal constraint severely restricts the spatial beamforming flexibility. The disjoint subarrays lack the fully-connected degrees of freedom required to actively steer nulls and orthogonalize the 16-QAM streams, leading to massive inter-stream interference. While the actual achievable multiplexing gain is strictly bounded by this combination of channel rank and AoSA constraints [41], the proposed Deep CS framework minimizes this unavoidable interference far better than conventional methods. Its superior CSI allows the digital baseband equalizers to suppress cross-stream leakage as effectively as physically possible under the hardware constraints.

4.4.5 Performance Across Varying Propagation Distances

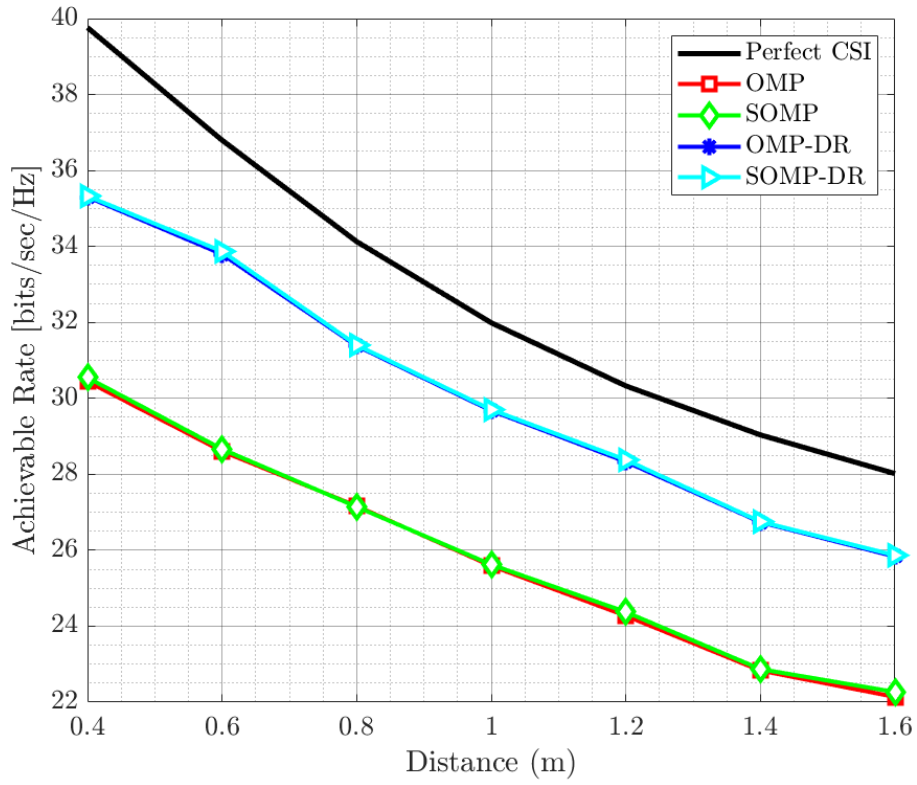
To provide a comprehensive evaluation of the system's spatial robustness, it is essential to analyze how the channel estimation algorithms perform as the physical distance between the transmitter and receiver increases. In the THz band, propagation distance is a critical factor due to the severe free-space path loss and molecular absorption, which rapidly degrade the received signal quality.

Figure 4.6 illustrates both the Normalized Mean Square Error (NMSE) and the Achievable Rate (AR) as a function of the transmission distance.

As observed in both subfigures, the performance of all evaluated conventional methods degrades as the distance increases. The NMSE (Figure 4.6a) steadily rises, and the Achievable Rate (Figure 4.6b) correspondingly decays. This overarching trend is primarily driven by the fundamental physics of the THz channel; as the distance extends, the received SNR inherently drops, making it increasingly difficult for the algorithms to extract the sparse multipath components from the noise floor. Furthermore, the changing distance fundamentally alters



(a) NMSE versus Distance



(b) Achievable Rate versus Distance

Figure 4.6: System performance evaluated across varying transmission distances.

the spherical wavefront curvature, complicating the near-field spatial signature.

A prominent feature of both plots is the perfect overlap between the standard algorithms (OMP, SOMP) and their DR counterparts (OMP-DR, SOMP-DR). This identical performance is a mathematically expected outcome. The DR technique operates by utilizing prior spatial information to extract a highly localized "sub-dictionary," effectively shrinking the search space for the sparse recovery algorithm. However, it does not alter the fundamental resolution, the basis vectors, or the grid structure of the dictionary itself. Consequently, the optimization algorithm selects the exact same basis atoms from the reduced dictionary as it would have from the full dictionary. Therefore, while DR massively reduces the computational complexity and search time, it yields the exact same mathematical estimation accuracy, resulting in perfectly overlapping curves.

Finally, it is necessary to address the intentional omission of the proposed DCS framework from this specific distance evaluation. The DCS architecture relies on a continuous generative prior that is explicitly optimized to understand the specific spatial manifold of the channel during the offline MAML training phase. Because near-field THz channel characteristics—particularly the spherical phase curvature—are highly sensitive to the exact physical distance, the continuous spatial distribution fundamentally shifts as the receiver moves. To accurately plot the DCS performance across a continuous sweeping distance, the framework would require the generation of a unique, massive channel dataset and a completely separate offline meta-training phase for every individual distance point on the x-axis. Executing this extensive retraining process for dozens of distinct geometric configurations is extremely time-intensive and computationally prohibitive, falling outside the scope of this specific spatial evaluation.

4.4.6 Computational Complexity and Online Inference Runtime

While the previous subsections established the superior estimation accuracy and SE of the proposed DCS framework, practical deployment in 6G Terahertz networks imposes strict latency constraints. Ultra-massive MIMO systems process an immense amount of spatial data, and the channel estimation algorithm must execute well within the coherence time to prevent critical delays. Therefore, evaluating the computational complexity and the actual online runtime is essential.

Table 4.3 summarizes the theoretical computational complexity of the evaluated algorithms alongside their exact numerical evaluations based on the simulation parameters. For the

standard dictionary-based methods (OMP and SOMP), the dominant computational burden arises from the massive, exhaustive correlation steps between the spatial dictionaries and the residual signals.

Table 4.3: Computational Complexity of Channel Estimation Algorithms

Algorithm	Computational Complexity (Big- O)	Numerical Value (FLOPs)	Relative to DCS
OMP	$O(K(LN_p G + L^3))$	335,552,320	5.120×
SOMP	$O(LKN_p G + KN_p L^2 + L^3)$	335,954,920	5.126×
OMP-DR	$O(K(LN_p G_{\text{OMP-DR}} + L^3))$	326,466,573	4.981×
SOMP-DR	$O(LKN_p G_{\text{SOMP-DR}} + KN_p L^2 + L^3)$	314,387,432	4.797×
Proposed DCS	$O(TN_s N_{p_{\text{DCS}}} N_t N_r)$	65,536,000	1.000×

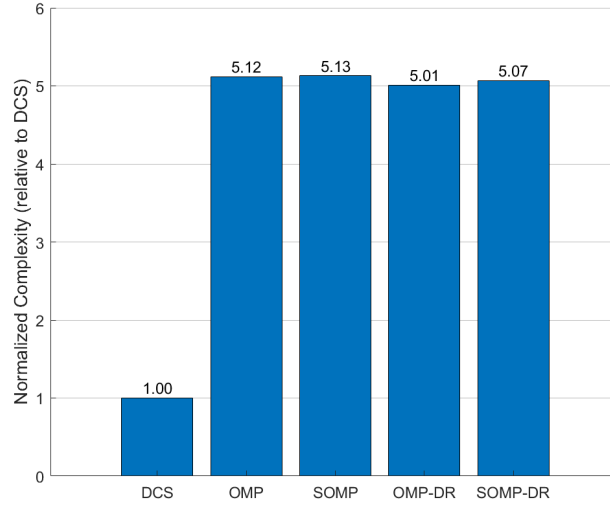
In contrast, the proposed DCS framework operates on a different computational paradigm. The immense mathematical burden of understanding the THz channel manifold is shifted entirely to the offline MAML training phase. During real-time online inference, the algorithm only requires T rapid gradient descent steps to optimize the low-dimensional latent vector, relying on highly parallelizable matrix multiplications rather than exhaustive grid searches.

Figure 4.7 validates this theoretical advantage by plotting the relative inference complexity and the physical average runtime measured during the simulations.

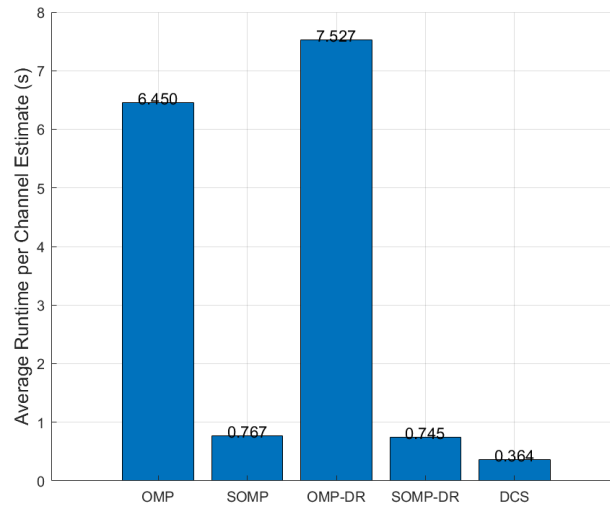
As illustrated in Figure 4.7a, standard OMP and SOMP exhibit extreme computational overhead, demanding over $5.1\times$ more operations than the DCS model. This mathematical complexity directly translates to physical latency, as demonstrated in Figure 4.7b. The conventional standard algorithms require massive amounts of time to estimate a single channel realization due to their unoptimized search grids. By bypassing the discrete dictionary entirely, the proposed DCS framework ensures an ultra-low latency response, confirming its viability for real-time deployment.

To address the severe latency of these standard baselines, DR techniques are often employed as a partial mitigation strategy. Figure 4.8 isolates the specific percentage of complexity reduction achieved by extracting highly localized sub-dictionaries prior to the grid search. As dictated by the parameters in Table 4.3, reducing the full dictionary size (G) to the localized sub-dictionaries ($G_{\text{SOMP-DR}}$) yields a direct proportional decrease in floating-point operations.

Although this DR technique provides a measurable optimization that reduces the relative complexity from $5.12\times$ to approximately $4.8\times$ (as seen in Figure 4.7a) and visible cuts in runtime execution (Figure 4.7b) it serves merely as a computational shortcut rather than



(a) Relative Inference Complexity



(b) Average Online Runtime (Seconds)

Figure 4.7: Overall computational efficiency evaluation. (a) Number of floating-point operations required relative to the proposed DCS baseline. (b) Absolute execution time per channel realization.

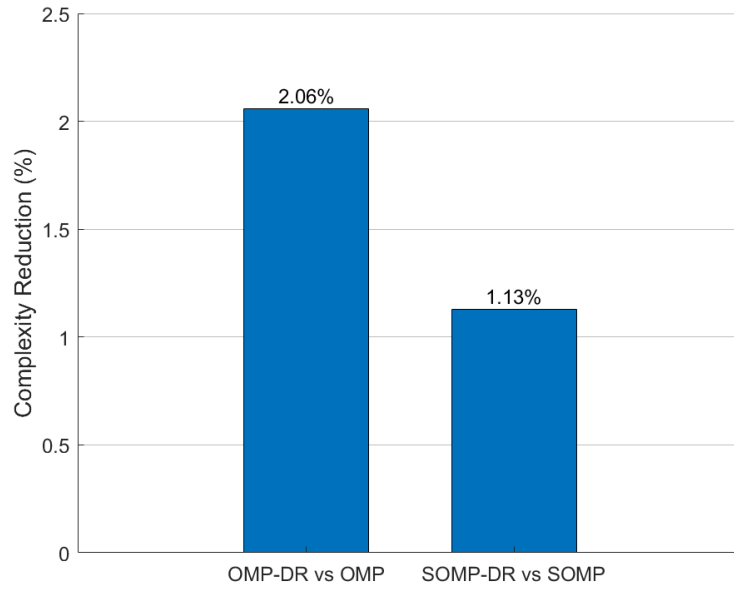


Figure 4.8: Percentage of computational complexity reduction achieved by incorporating the DR technique into standard OMP and SOMP algorithms.

a fundamental solution. Its continued reliance on a mathematically heavy sparse-recovery paradigm prevents it from bridging the massive latency gap established by the Deep CS framework.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis investigated the viability of DCS for near-field channel estimation in THz Ultra-Massive MIMO systems. Operating under the strict hardware constraints of a sub-connected AoSA architecture.

- The project developed a comprehensive framework to compare a learned generative-prior approach against conventional dictionary-based algorithms, including OMP, SOMP, and their Dictionary Reduction variants.
- The evaluation demonstrated that the proposed DCS framework provides an estimation advantage over classical methods. By learning the underlying THz channel manifold from data, DCS bypasses the grid-mismatch and discretization errors that heavily bottleneck conventional sparse recovery. This results in a lower Normalized Mean Square Error across a wide range of signal-to-noise ratios.
- The DCS model reliably recovers high-quality channel state information using significantly fewer pilot measurements, directly increasing the effective achievable rate by freeing up the coherence block for payload transmission.
- The end-to-end Bit Error Rate analysis further highlighted both the strengths of the framework and the physical realities of the system. Although DCS maintained superior reliability in single-stream transmission, the multi-stream evaluation revealed a universal performance degradation across all algorithms. This confirmed that while DCS optimally

minimizes interference, spatial multiplexing remains fundamentally limited by the physical sparsity of the THz environment and the block-diagonal structural constraints of the AoSA hardware.

- The computational analysis proved the practical deployment viability of the learned estimator. Conventional sparse recovery methods demand computationally heavy, exhaustive grid searches during real-time operation. Even with the integration of DR techniques, which shrink the search space to provide a partial mitigation, the baseline methods still suffer from high latency. In contrast, DCS shifts the heavy mathematical burden entirely to the offline meta-learning phase, reducing online inference to rapid latent optimization. This ensures an ultra-low latency response that easily meets the strict coherence time limits of next-generation networks.

5.2 Future Work

While the proposed framework establishes a strong foundation, several critical directions remain open to bridge the gap between synthetic simulations and real-world 6G deployment.

- The first suggestion is the development of **geometry-aware generalization**. Because near-field spatial signatures are highly sensitive to propagation distance, future models should be distance-conditioned, allowing a single trained network to adapt to continuously varying spatial configurations without requiring distinct offline retraining phases. This could be seamlessly coupled with lightweight online adaptation mechanisms to rapidly adjust the learned prior to dynamic environments, blockages, or changing atmospheric conditions.
- Another highly promising direction is expanding the system model to address the complex physical phenomena of wideband THz channels. Given massive bandwidths approaching 10 GHz, the physical propagation delay across the ultra-massive array causes the beam to shift across subcarriers, a severe complication known as **beam squint**. Future iterations of the DCS framework must integrate frequency-aware latent dimensions to natively track and compensate for this spatial-wideband effect. Furthermore, the architecture should be upgraded to handle cross-field propagation,

automatically distinguishing and estimating mixed-field channels where near-field Line-of-Sight paths and far-field scattered components coexist simultaneously.

- Finally, practical validation and offline efficiency must be addressed. Future studies should evaluate the DCS framework under realistic RF hardware **non-idealities and experimental** measured channel data to confirm its robustness. Concurrently, investigating model compression, quantized inference, and lighter generator architectures will be vital to reduce the heavy computational demands of the offline training stage, moving the framework closer to practical implementation in resource-constrained environments.

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